Design and Evaluation of a 3D Cartographic Prototype for Visualizing Urban Thermal Stress at City-Block Scale

Yi-Ru (Kelly) Liang

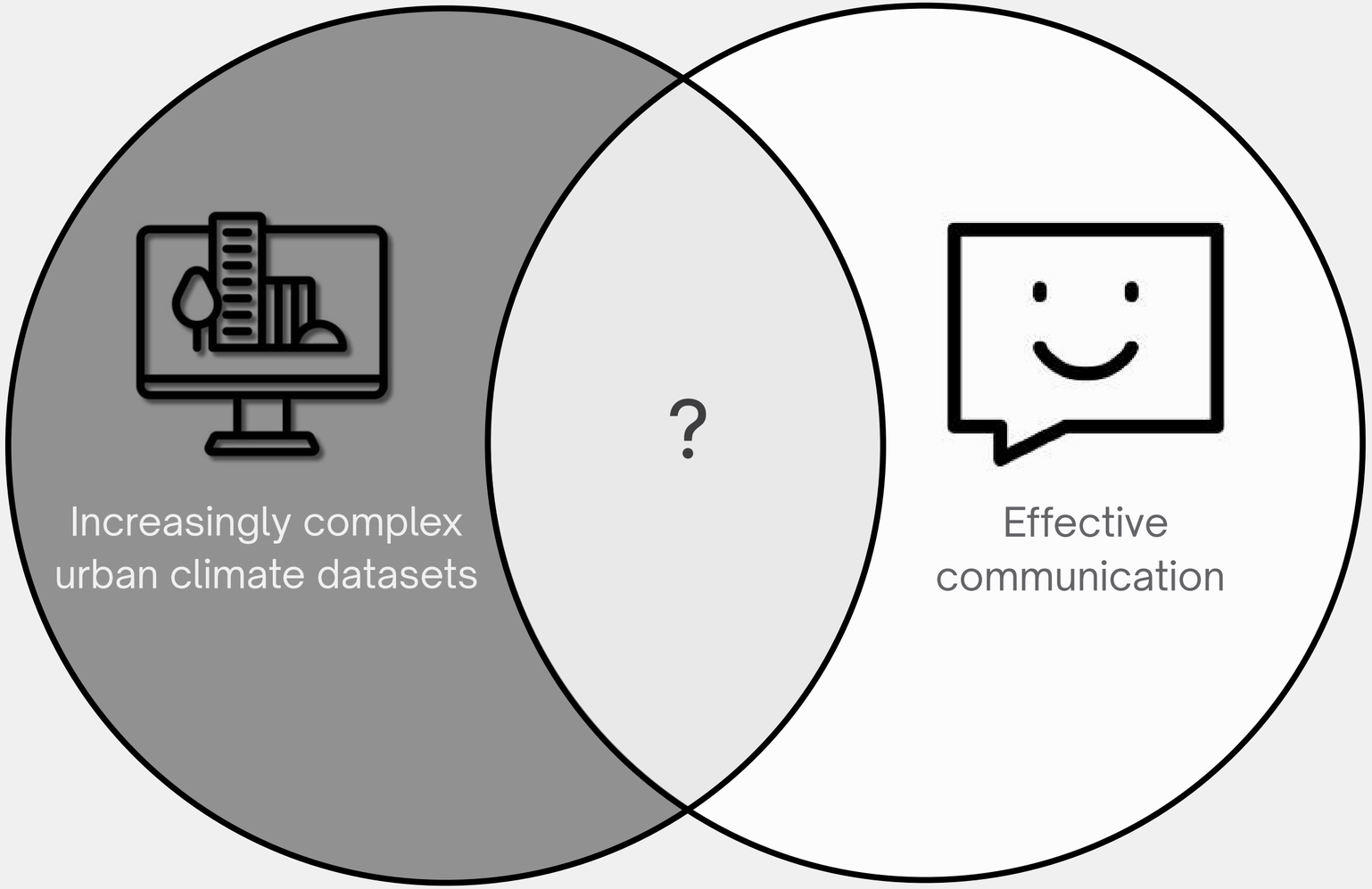
Thesis defense in M.Sc. Cartography,
Technische Universität Wien (TU Wien)

September 17th, 2026

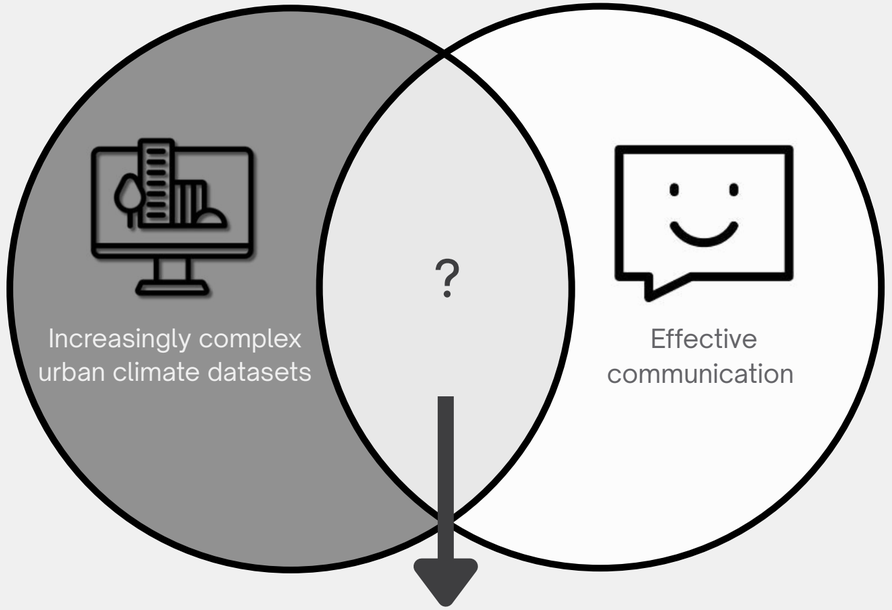
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Chair Professor: Univ.Prof. Mag.rer.nat. Dr.rer.nat Georg Gartner, TU Wien

Motivation 3
Research Objectives and Questions 4
Dataset and Simulation 7
Methodology 10
3D Prototype 11
User Study 22
Results and Discussion 27
Key Takeaways 37

MOTIVATION


Increasingly complex
urban climate datasets
?
Effective
communication

RESEARCH OBJECTIVES



Design and evaluate a 3D visualization prototype to support the exploration and interpretation of urban climate data.

RESEARCH QUESTIONS

3D Visualization

O1: suitable 3D visual encodings and occlusion mitigation

RQ 1.1: How do **3D visual encodings** influence the interpretation?

RQ 1.2: How effectively do **opacity controls** mitigate occlusions?

O2: Interactive controls design

RQ 2.1: In what ways do **interface controls** support identification and comparison tasks?

RQ 2.2: How does the integration of **activity-based settings** affect the interpretation?

RESEARCH QUESTIONS

Evaluation

O3: Evaluate the influence of user traits, prior knowledge, and user behaviors

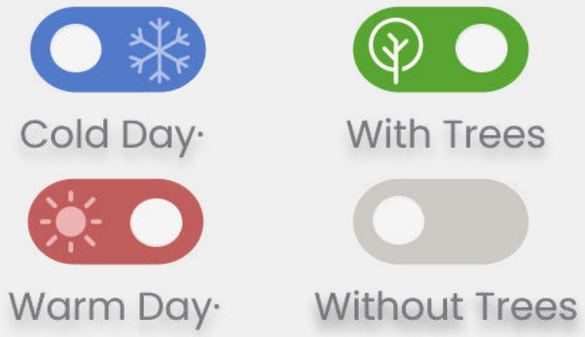
RQ 3.1: How do **user factors** influence how users explore a 3D environment?

RQ 3.2: What 3D visualization states or interaction patterns trigger **user frustration**?

DATASET AND SIMULATION

The **Universal Thermal Climate Index (UTCI)**, an indicator of urban thermal stress, is used as a case study.

A pre-computed UTCI dataset across **six height levels from 0 to 10 meters**, representing pedestrian exposure up to average tree canopy height

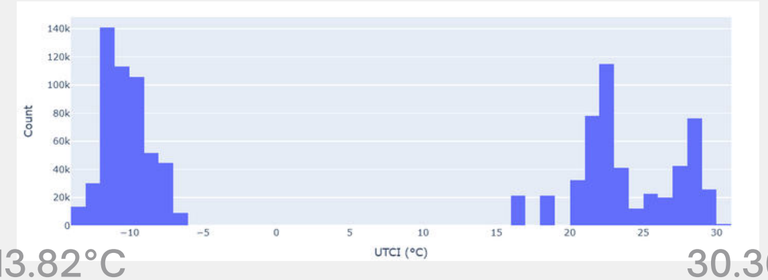


4 simulated scenarios

UTCI (°c) range	Stress category
Above +46	Extreme heat stress
+38 to +46	Very strong heat stress
+32 to +38	Strong heat stress
+26 to +32	Moderate heat stress
+9 to +26	No thermal stress
+9 to 0	Slight cold stress
0 to -13	Moderate cold stress
-13 to -27	Strong cold stress
-27 to -40	Very strong cold stress
Below -40	Extreme cold stress

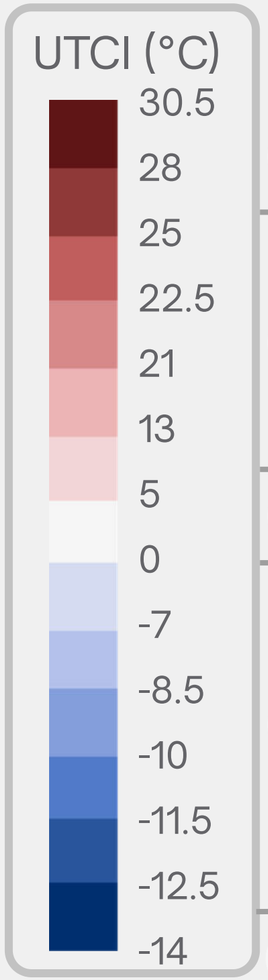
DATASET AND SIMULATION

UTCI dataset



-13.82°C 30.36°C

UTCI data distribution



Moderate heat stress

26°C

No thermal stress

9°C

Slight cold stress

0°C

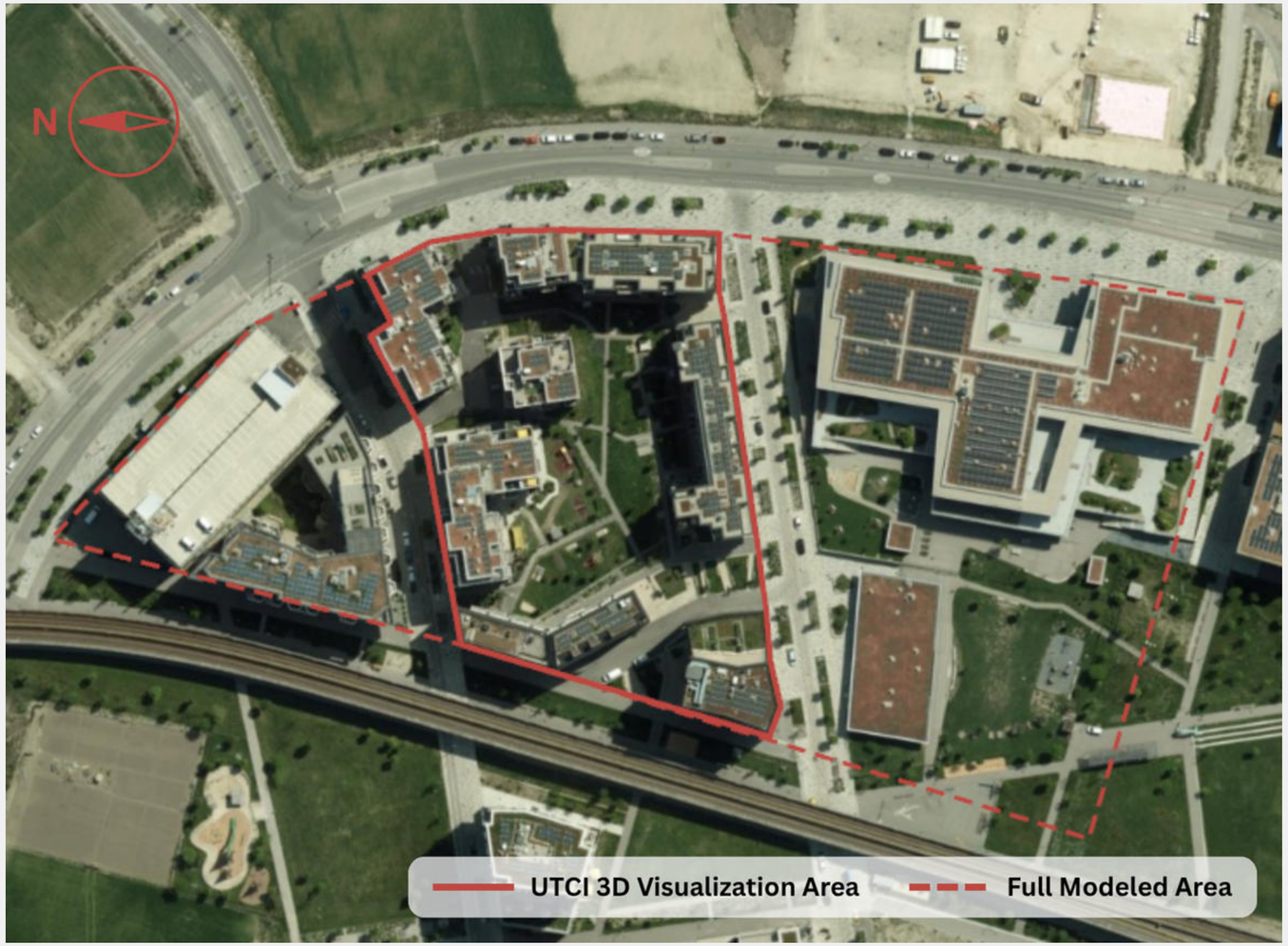
Moderate cold stress

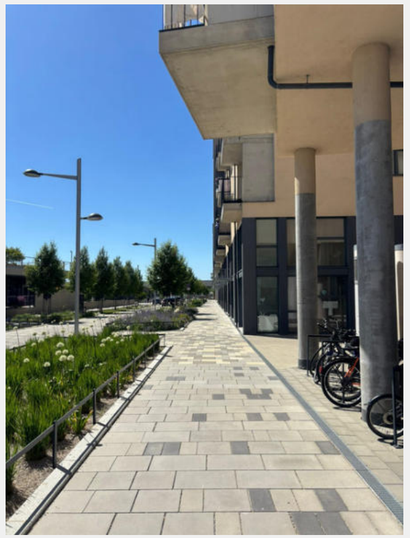
-13°C

Strong cold stress

UTCI (°c) range	Stress category
Above +46	Extreme heat stress
+38 to +46	Very strong heat stress
+32 to +38	Strong heat stress
+26 to +32	Moderate heat stress
+9 to +26	No thermal stress
+9 to 0	Slight cold stress
0 to -13	Moderate cold stress
-13 to -27	Strong cold stress
-27 to -40	Very strong cold stress
Below -40	Extreme cold stress

STUDY AREA







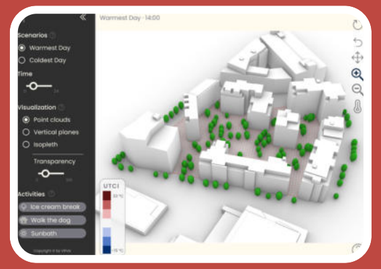


METHODOLOGY

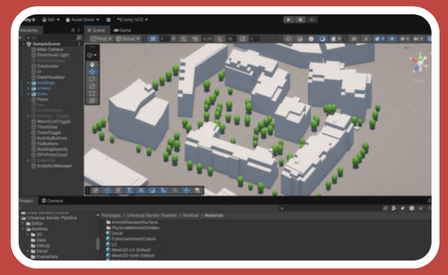
Exploration, Design, and Implementation



Literature review



Mock-ups



Implementation



Prototype

Empirical Evaluation and Analysis

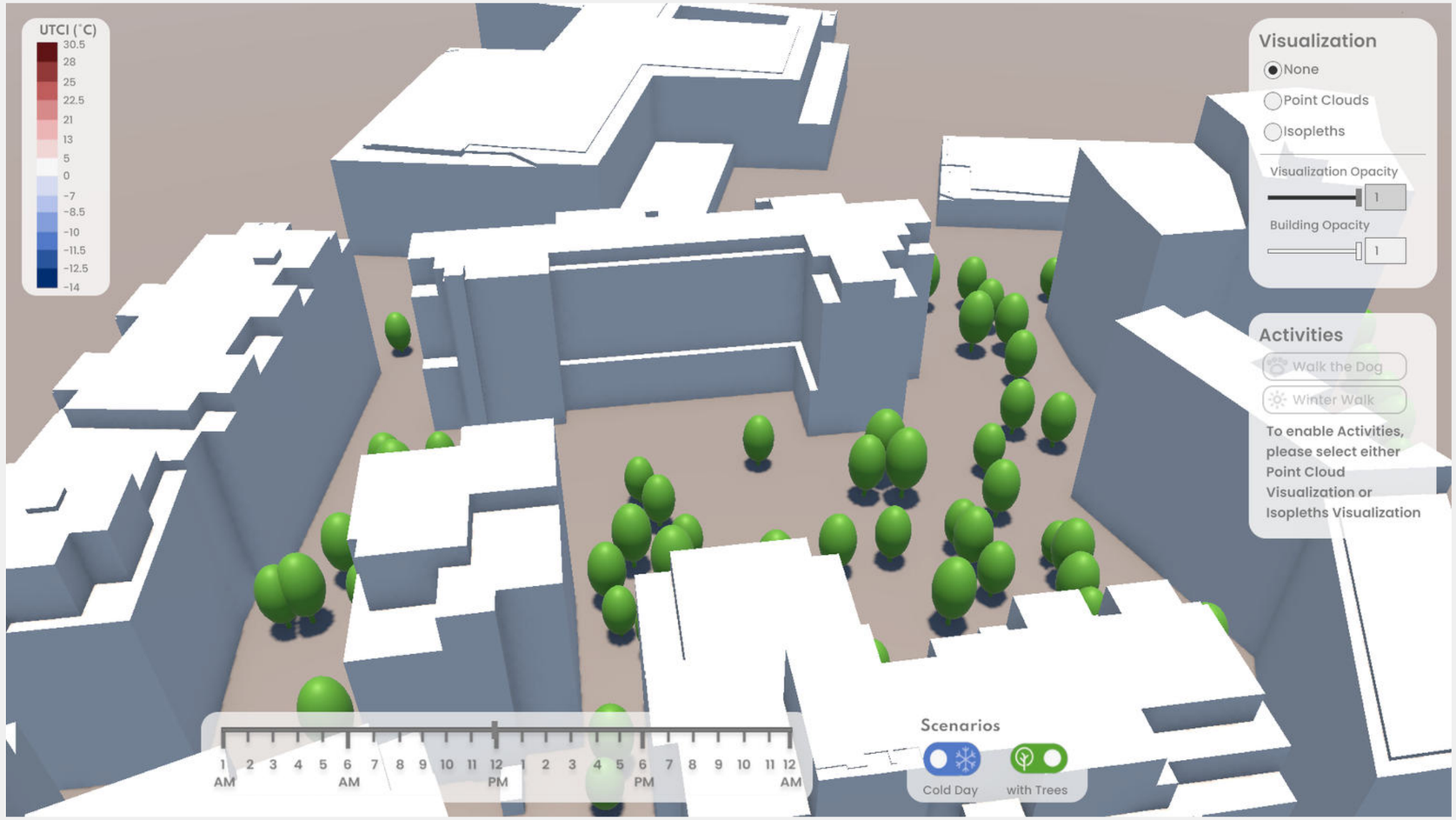


User study



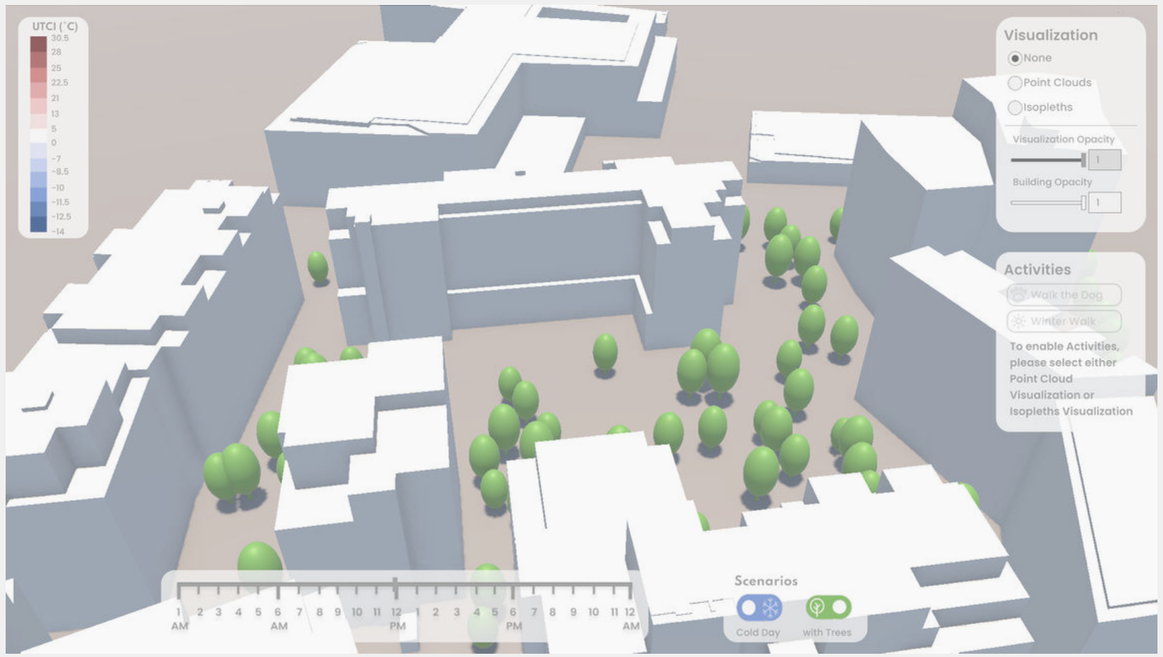
Analysis of results

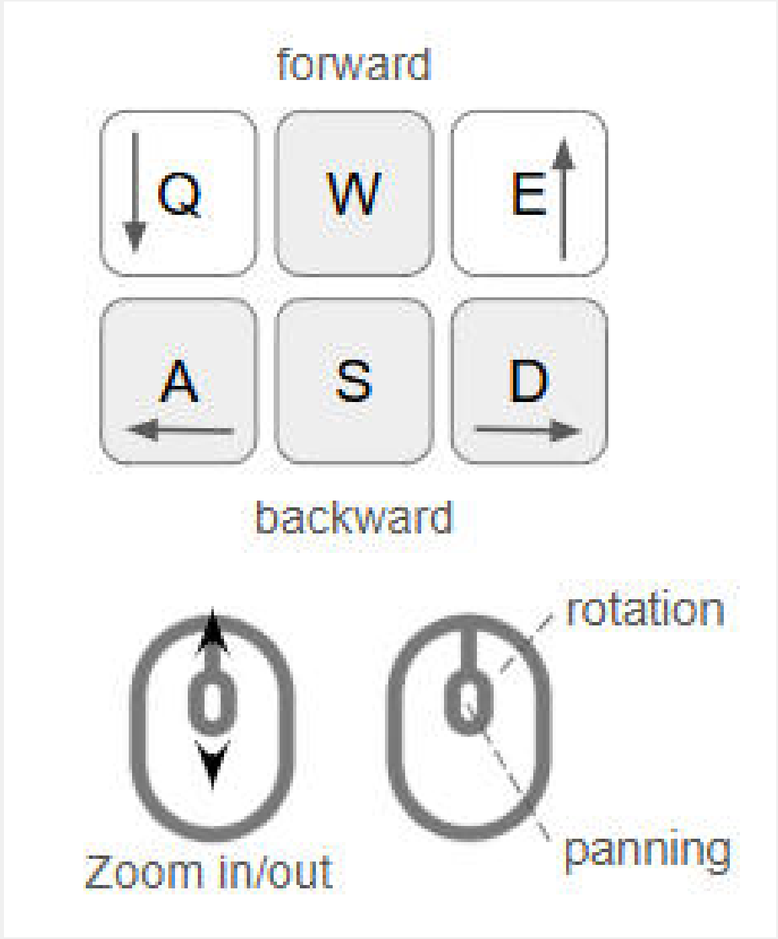
3D PROTOTYPE



Initial screen displayed when launching the prototype

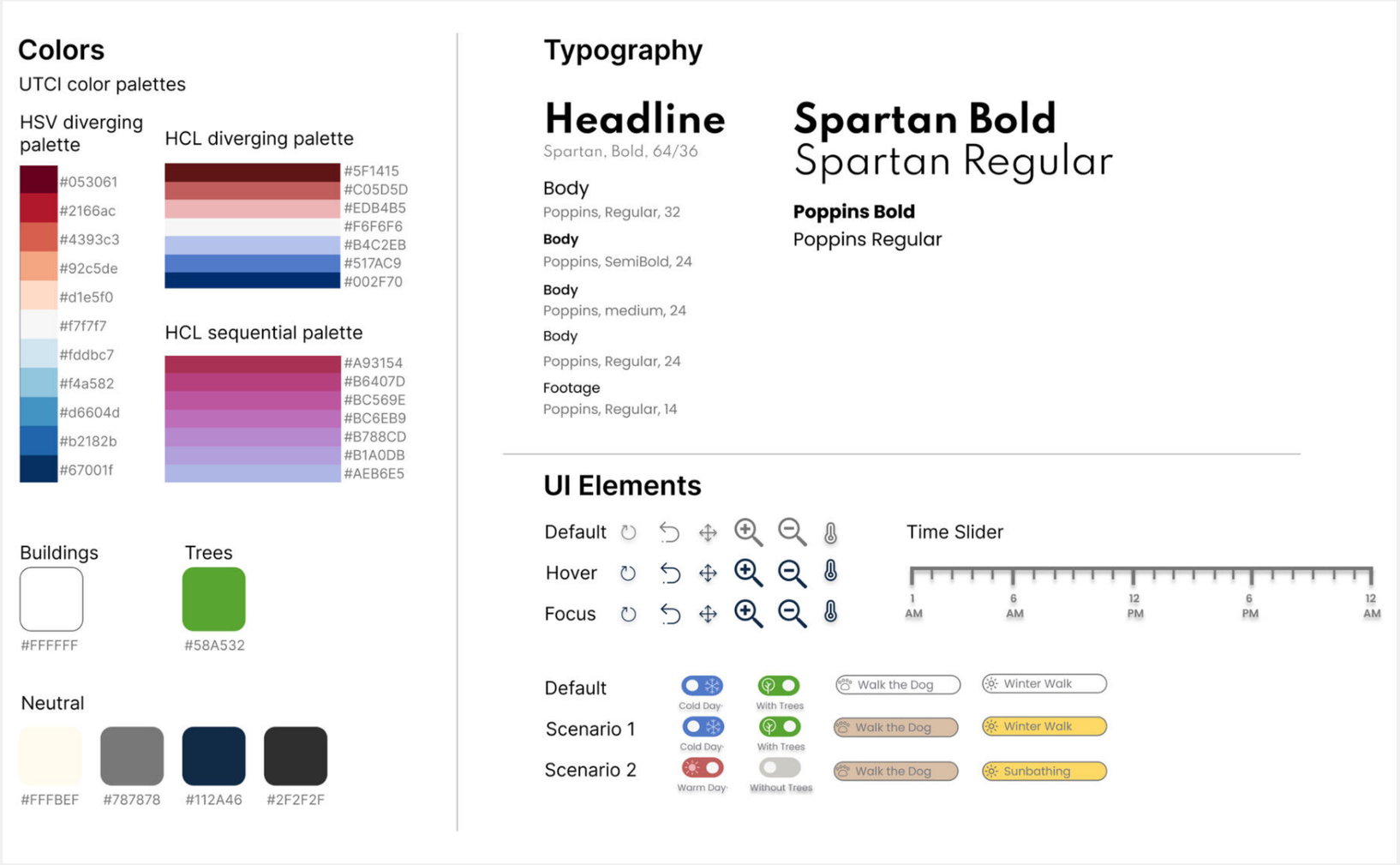
3D PROTOTYPE



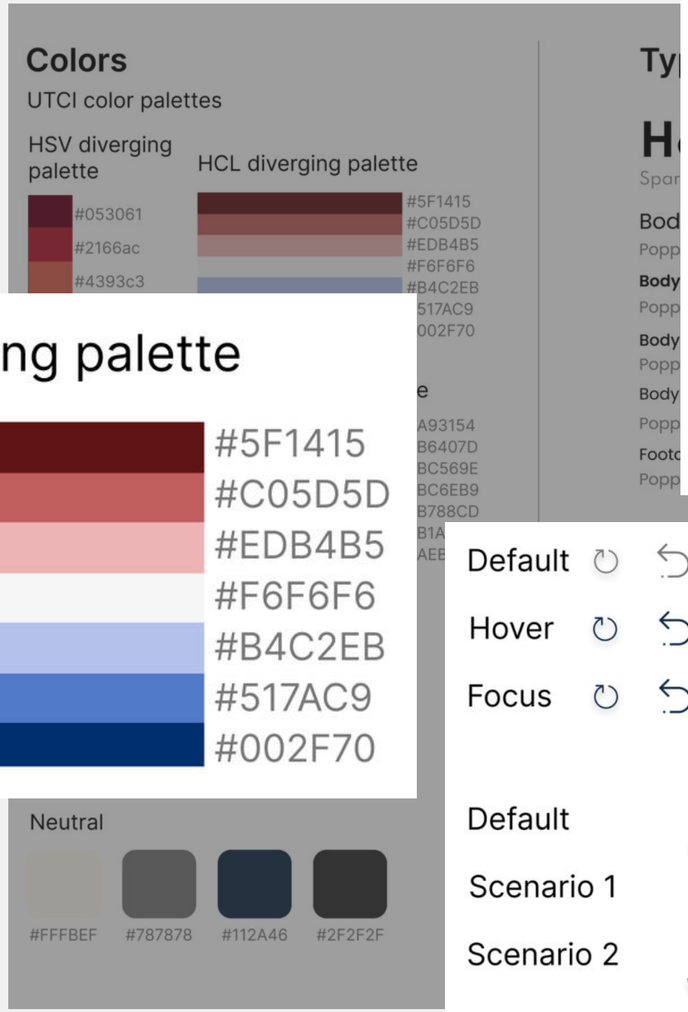


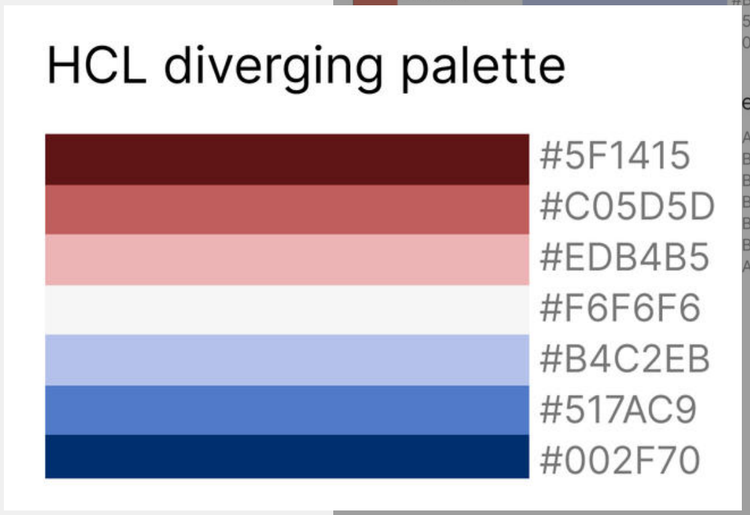
3D navigation reference guide

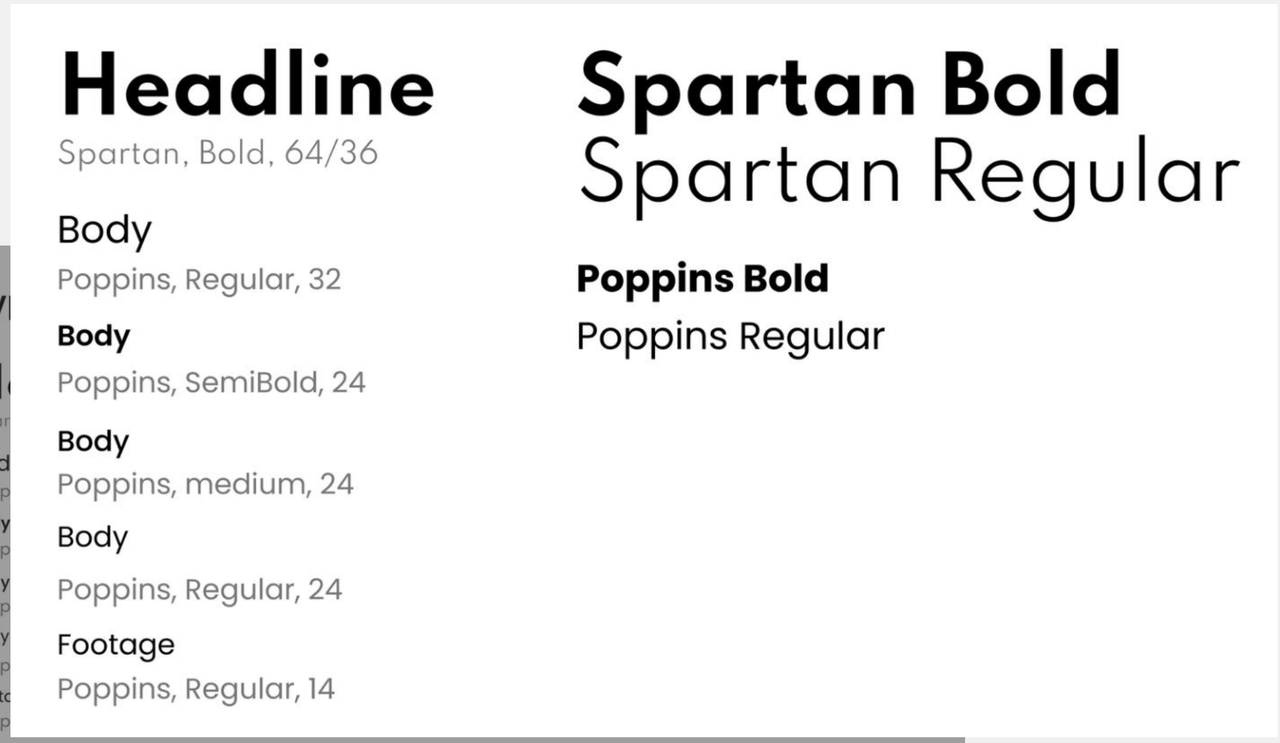
3D PROTOTYPE



3D PROTOTYPE



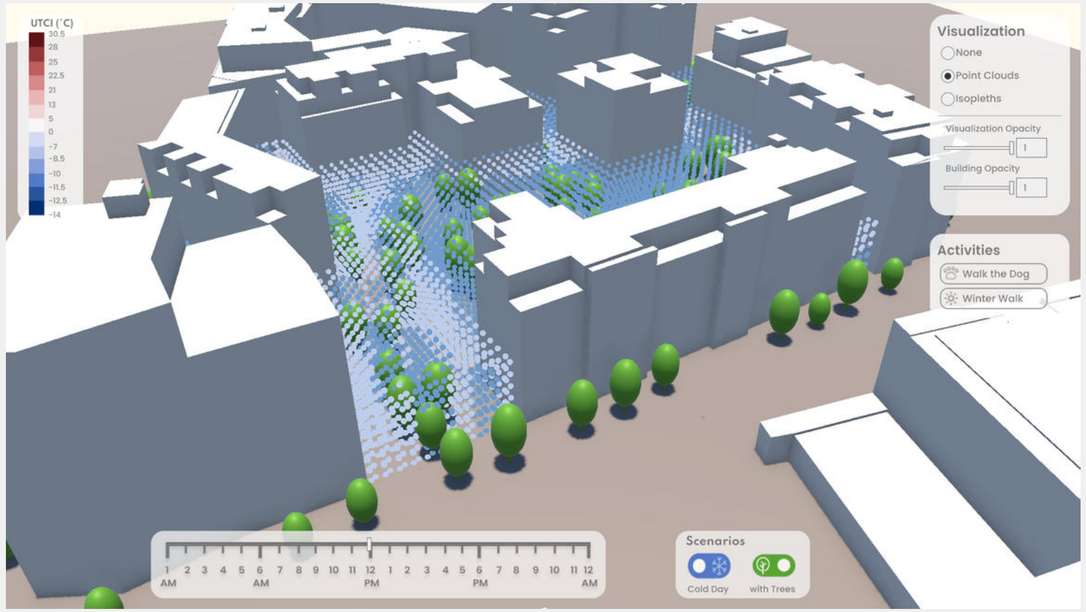




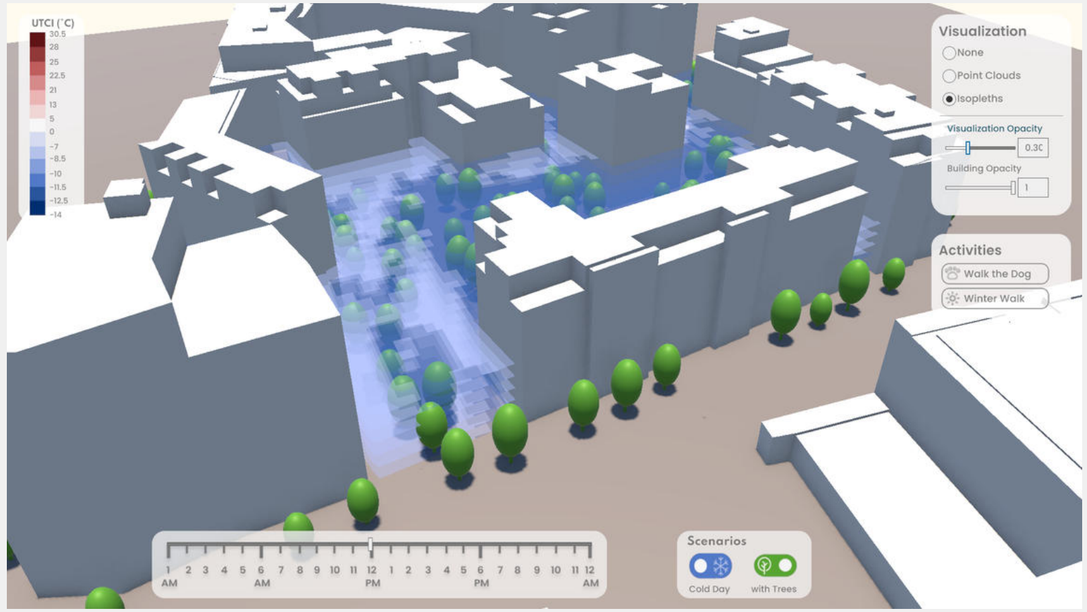


3D PROTOTYPE

3D visual encodings



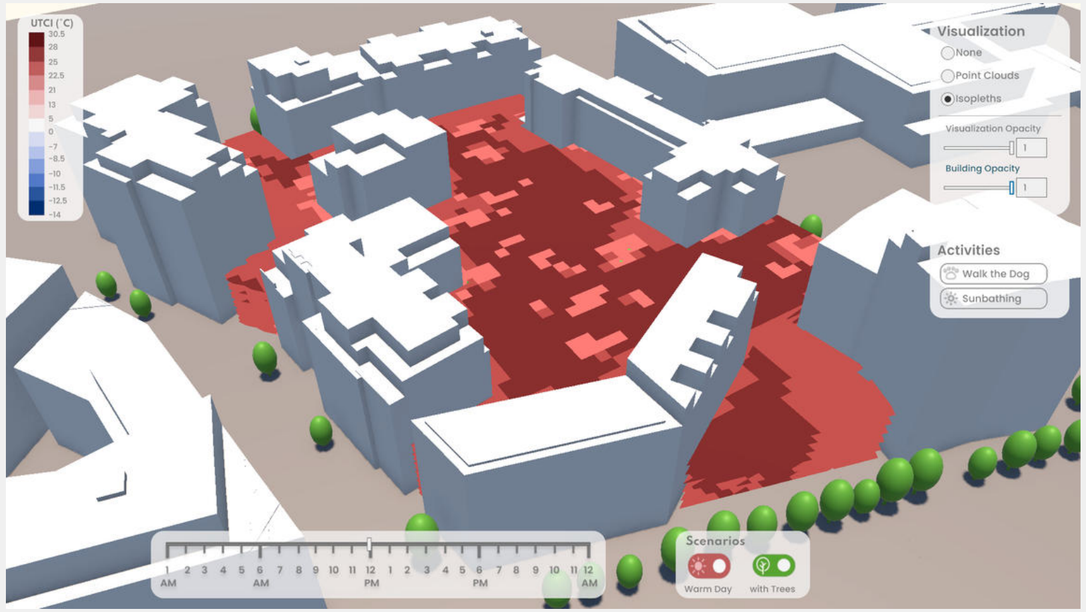
Point clouds (Gautier et al., 2020)



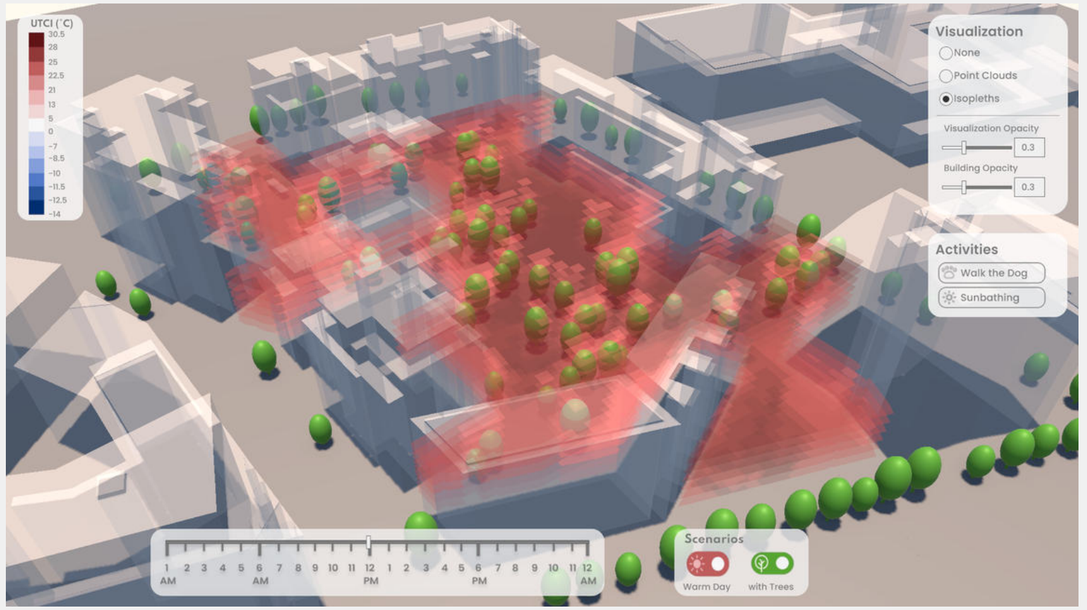
Horizontal planes with 30% transparency (Gautier et al., 2020)

3D PROTOTYPE

Opacity sliders



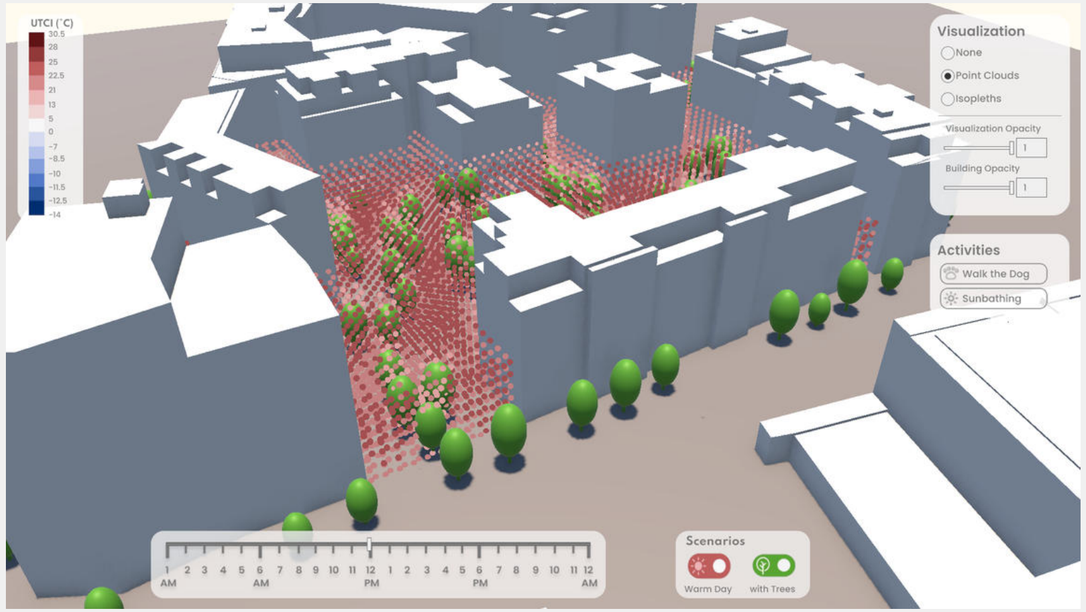
Before adjusting transparency



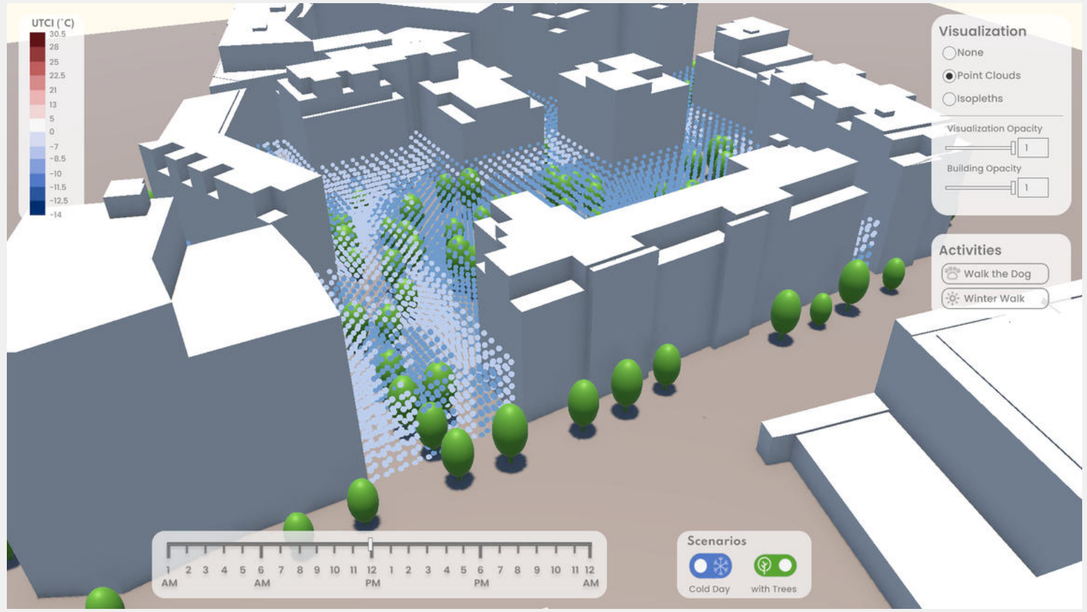
After adjusting both visulaization and building transparency

3D PROTOTYPE

Scenario toggles - Warm/Cold condition



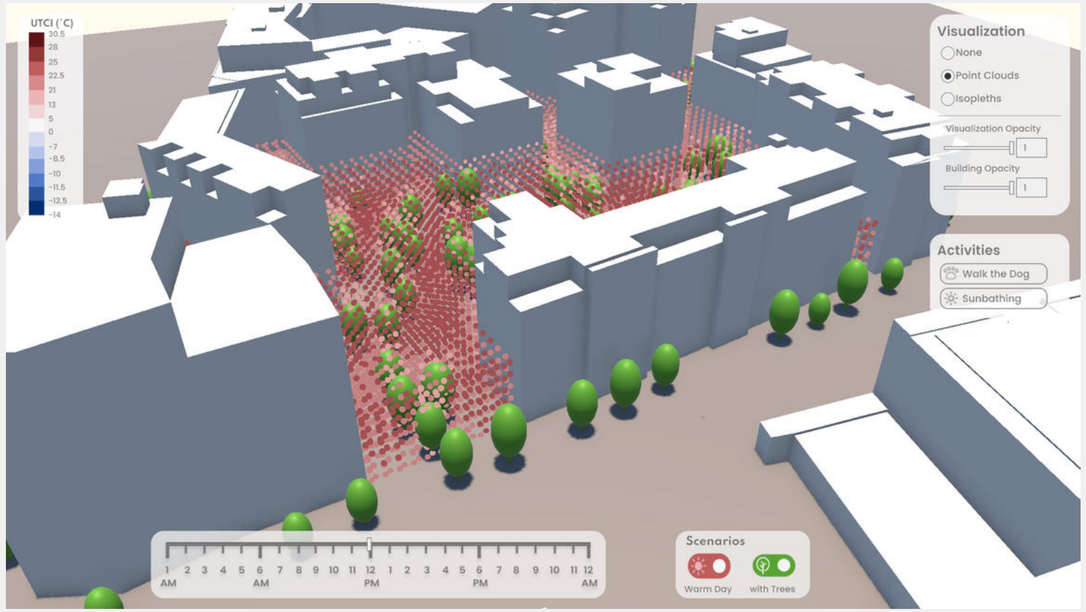
Warm Day



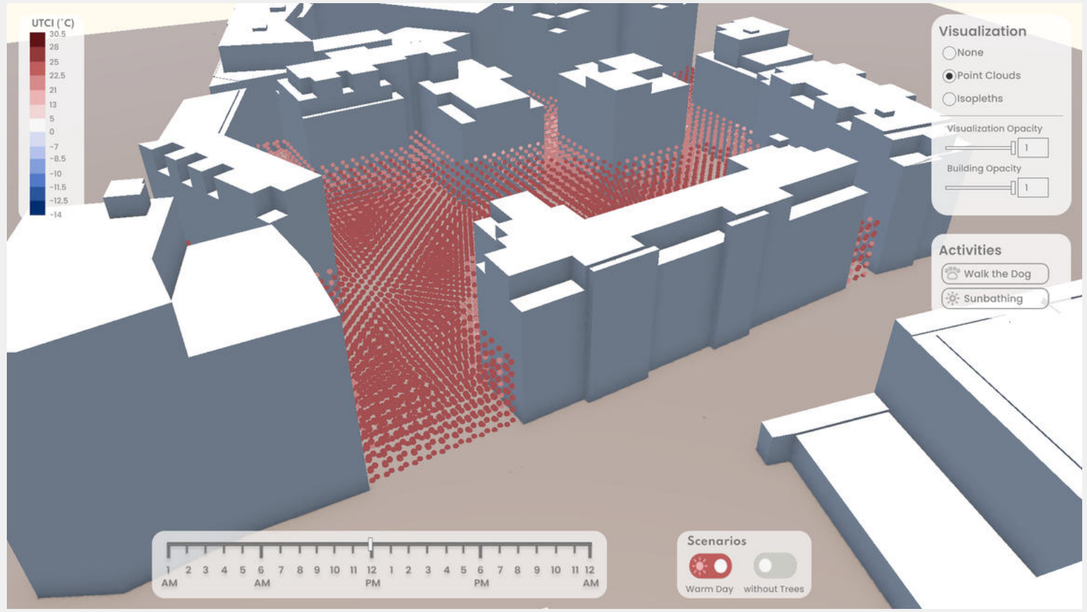
Cold Day

3D PROTOTYPE

Scenario toggles - Tree condition



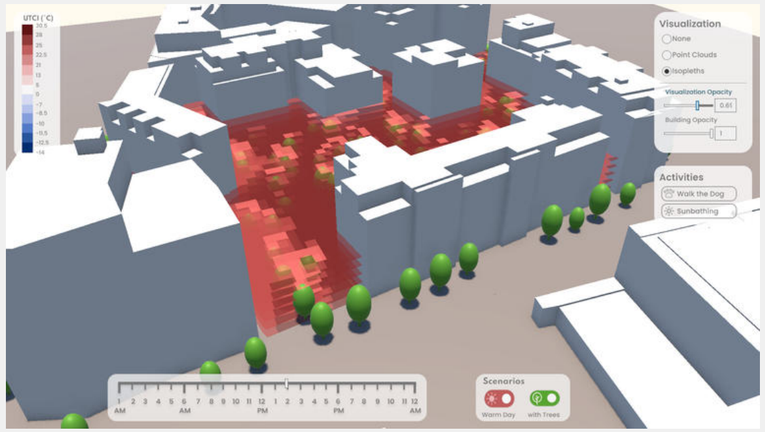
With trees



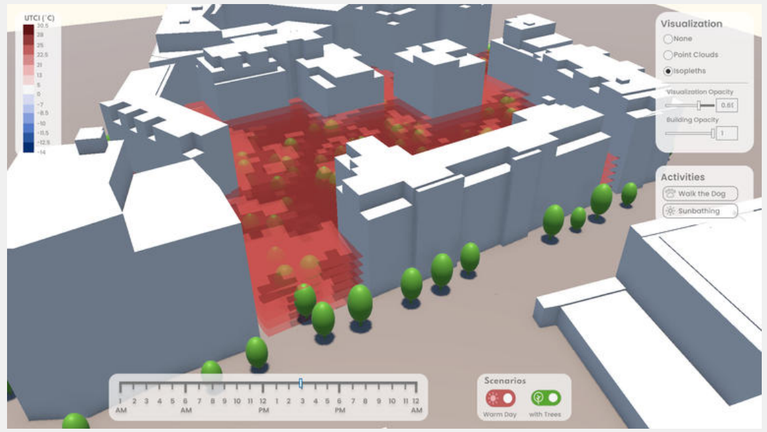
Without trees

3D PROTOTYPE

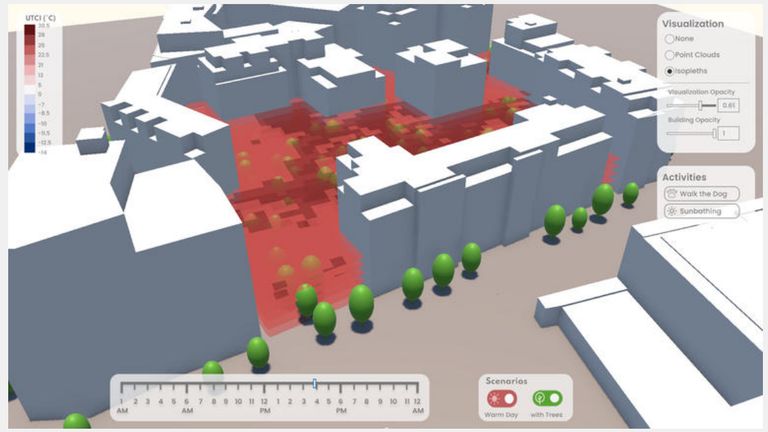
Time slider



2:00 PM



3:00 PM



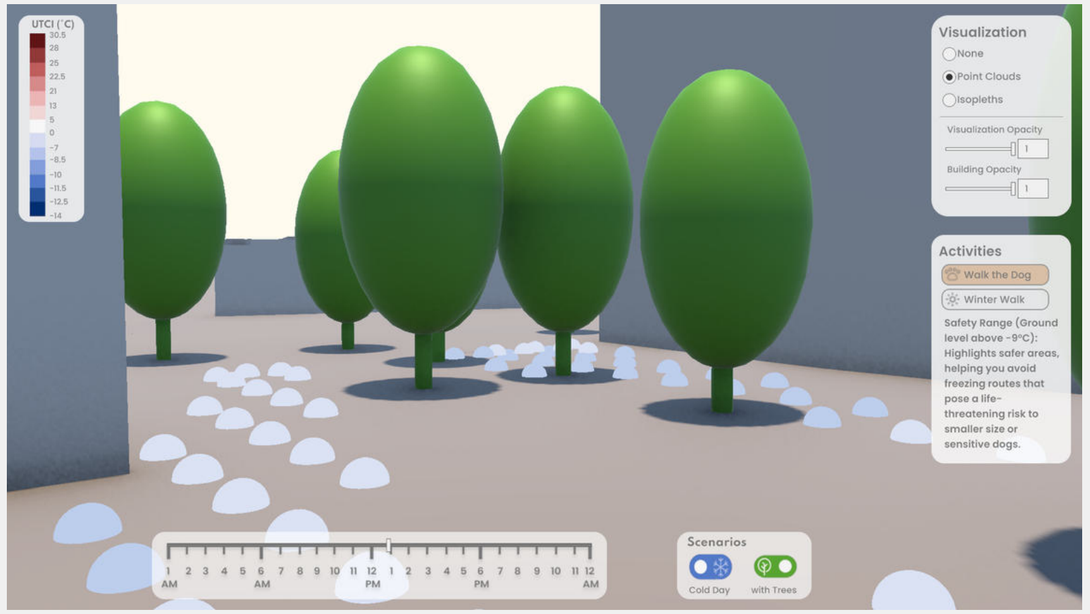
4:00 PM

3D PROTOTYPE

Activity mode



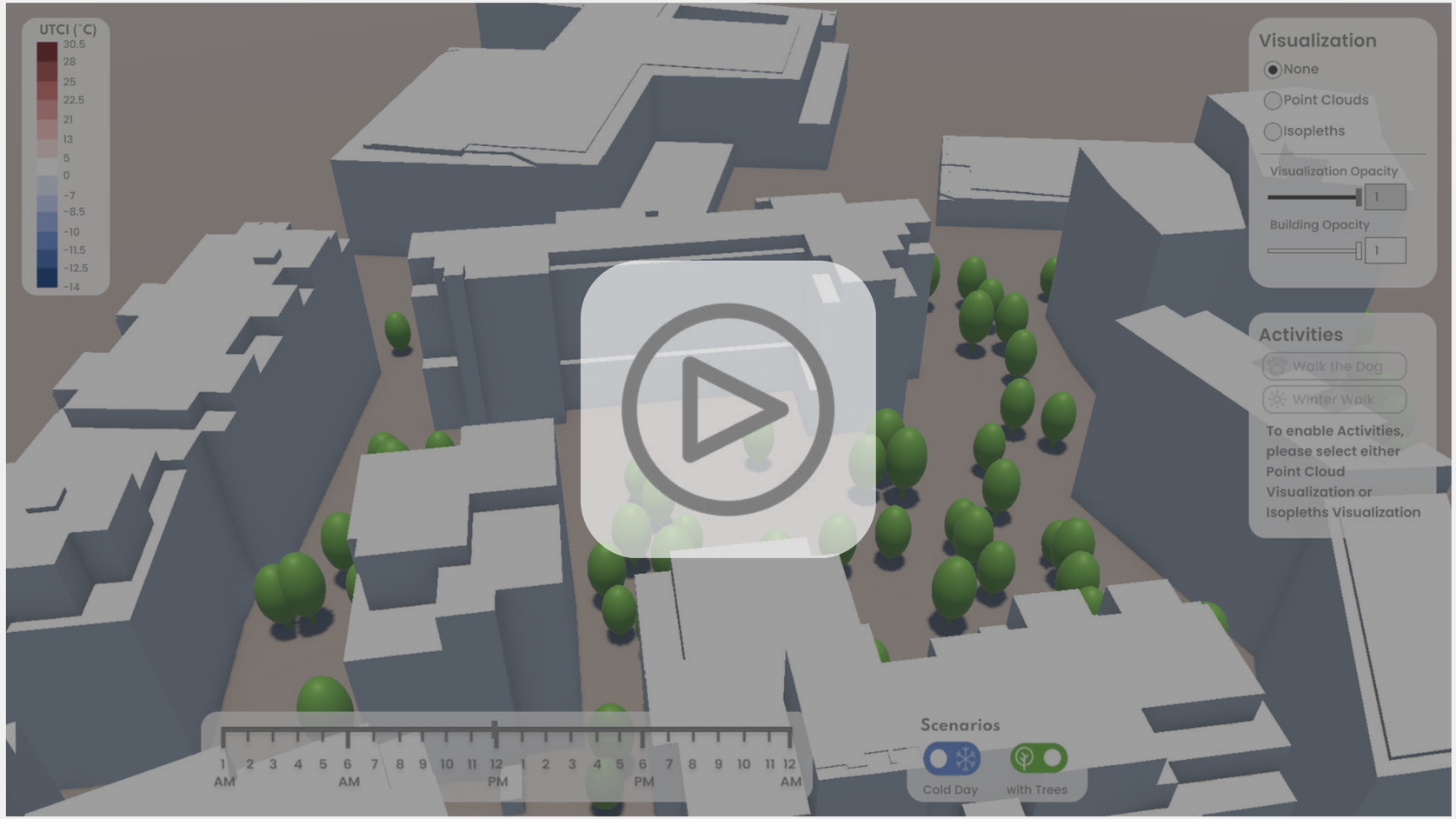
Sunbathing / Winter Walk - Isolates comfortable areas for solar exposure



Walk the Dog - Isolates ground-level thermal safety areas for dogs

3D PROTOTYPE

Demo video



USER STUDY

Participants

- Aged between 18 to over 65
- 10 male, 10 female



The Locals
(n=10)

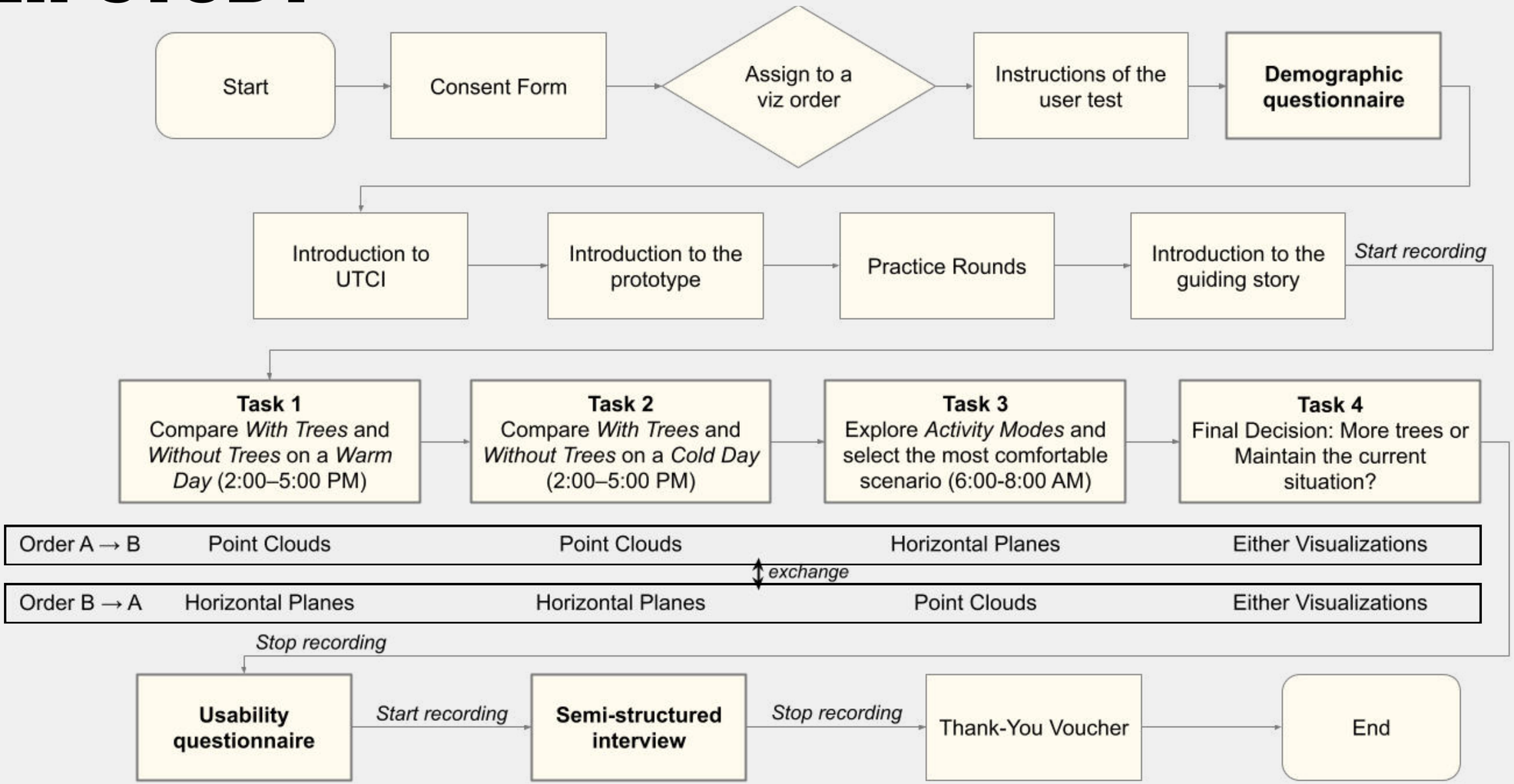
- People who spend a significant amount of time in the study area.
- Potential end users



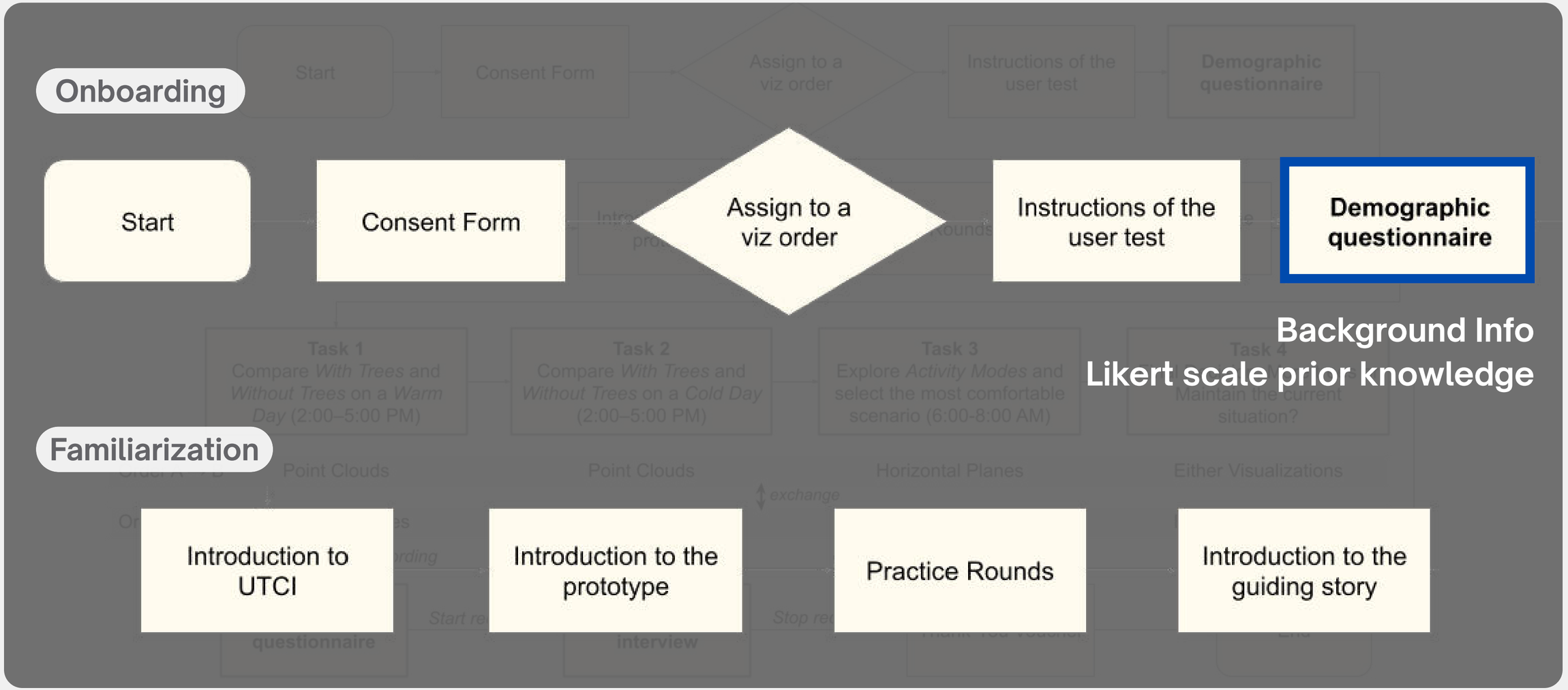
The Experts
(n=10)

- Domain experts in cartography, geovisualization, or scientific visualization
- Critical assessment of the prototype

USER STUDY


Start
Consent Form
Assign to a viz order
Instructions of the user test
Demographic questionnaire
Introduction to UTCI
Introduction to the prototype
Practice Rounds
Introduction to the guiding story
Start recording
Task 1
Compare With Trees and Without Trees on a Warm Day (2:00–5:00 PM)
Task 2
Compare With Trees and Without Trees on a Cold Day (2:00–5:00 PM)
Task 3
Explore Activity Modes and select the most comfortable scenario (6:00-8:00 AM)
Task 4
Final Decision: More trees or Maintain the current situation?
Order A → B
Point Clouds
Point Clouds
Horizontal Planes
Either Visualizations
exchange
Order B → A
Horizontal Planes
Horizontal Planes
Point Clouds
Either Visualizations
Stop recording
Usability questionnaire
Start recording
Semi-structured interview
Stop recording
Thank-You Voucher
End

USER STUDY


Onboarding
Start
Consent Form
Assign to a viz order
Instructions of the user test
Demographic questionnaire
Background Info
Likert scale prior knowledge
Familiarization
Introduction to UTCI
Introduction to the prototype
Practice Rounds
Introduction to the guiding story

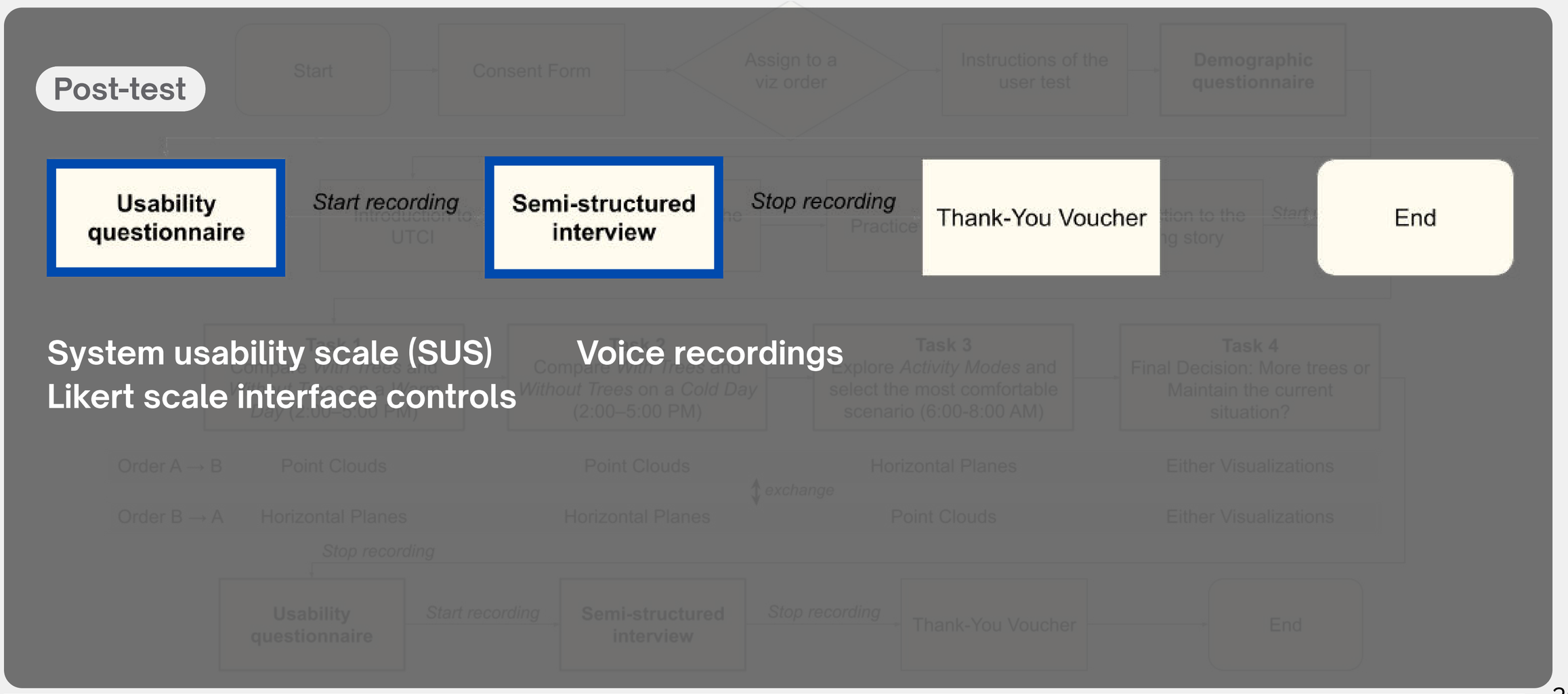
USER STUDY

Exploration tasks

	Task 1	Task 2	Task 3	Task 4
	Compare *With Trees* and *Without Trees* on a *Warm Day* (2:00–5:00 PM)	Compare *With Trees* and *Without Trees* on a *Cold Day* (2:00–5:00 PM)	Explore *Activity Modes* and select the most comfortable scenario (6:00-8:00 AM)	Final Decision: More trees or Maintain the current situation?
Order A → B	Point Clouds	Point Clouds	Horizontal Planes	Either Visualizations
		exchange		
Order B → A	Horizontal Planes	Horizontal Planes	Point Clouds	Either Visualizations

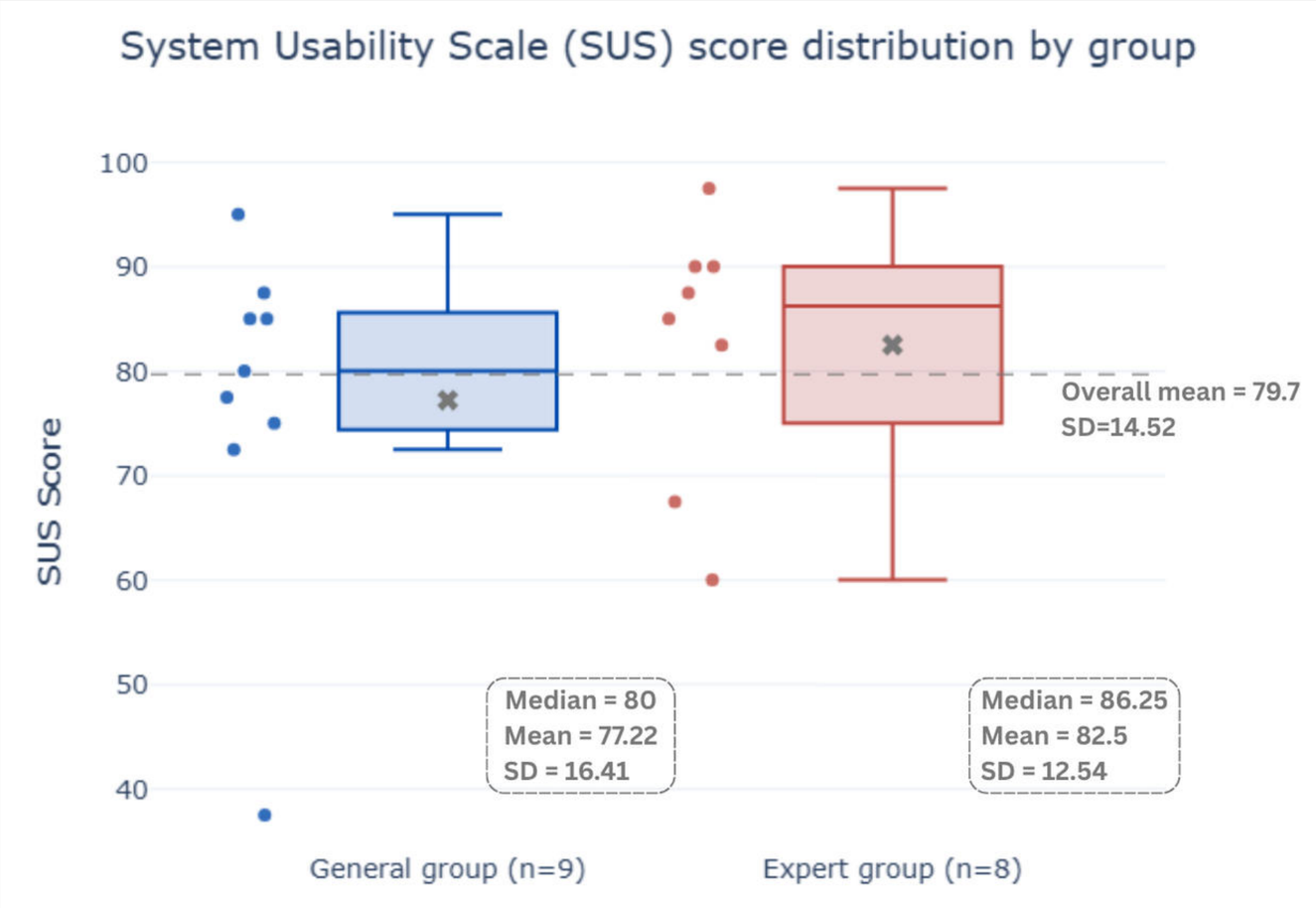
Screen and voice recordings
Interaction logs
Think-aloud protocols

USER STUDY


Post-test
Usability questionnaire
Start recording
Semi-structured interview
Stop recording
Thank-You Voucher
End
System usability scale (SUS)
Likert scale interface controls
Voice recordings

RESULTS AND DISCUSSION

System usability scale (SUS)



RESULTS AND DISCUSSION

Interface controls and exploration behaviors



≈ 50%

On average, half of all interactions used the time slider
The most frequently used interface control



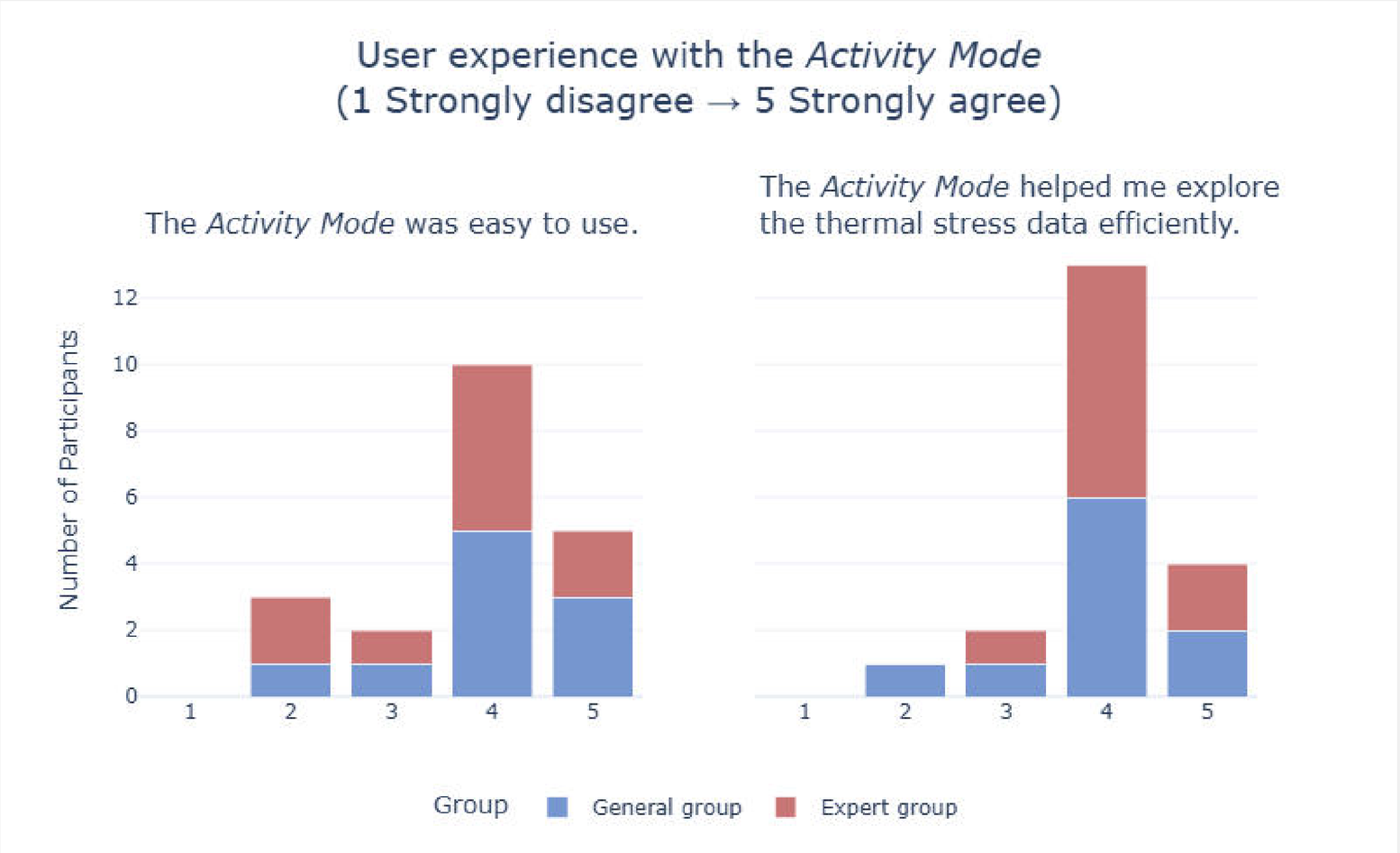
≈ 27%

On average, over a quarter of all interactions used the *Tree Condition* toggle
The second most frequently used control

RESULTS AND DISCUSSION

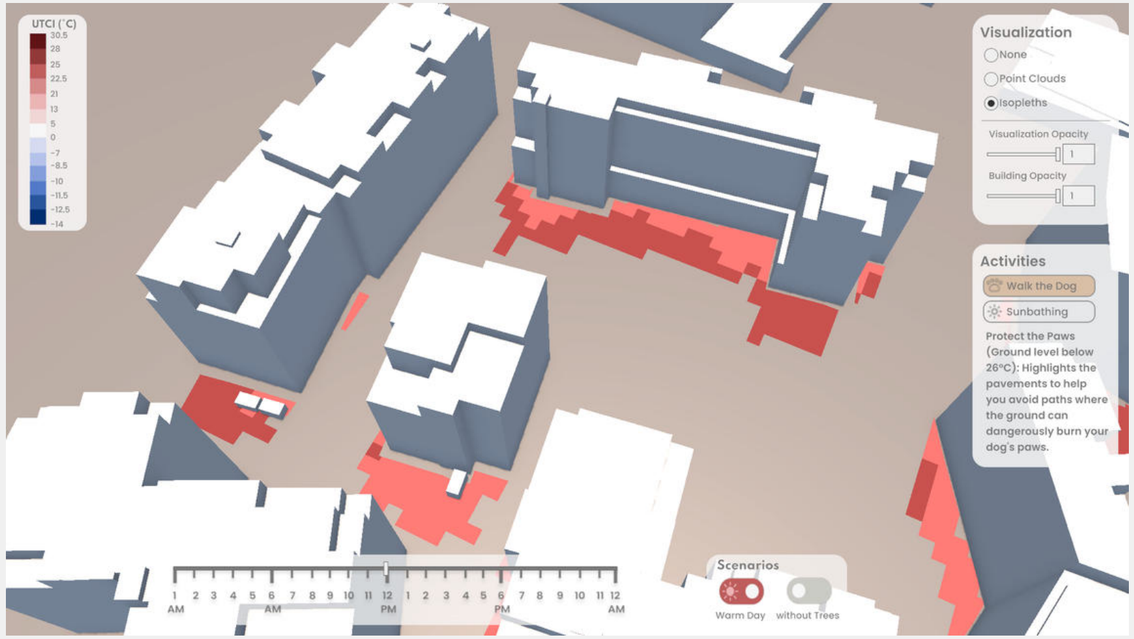
After understanding how it works...

Activity mode

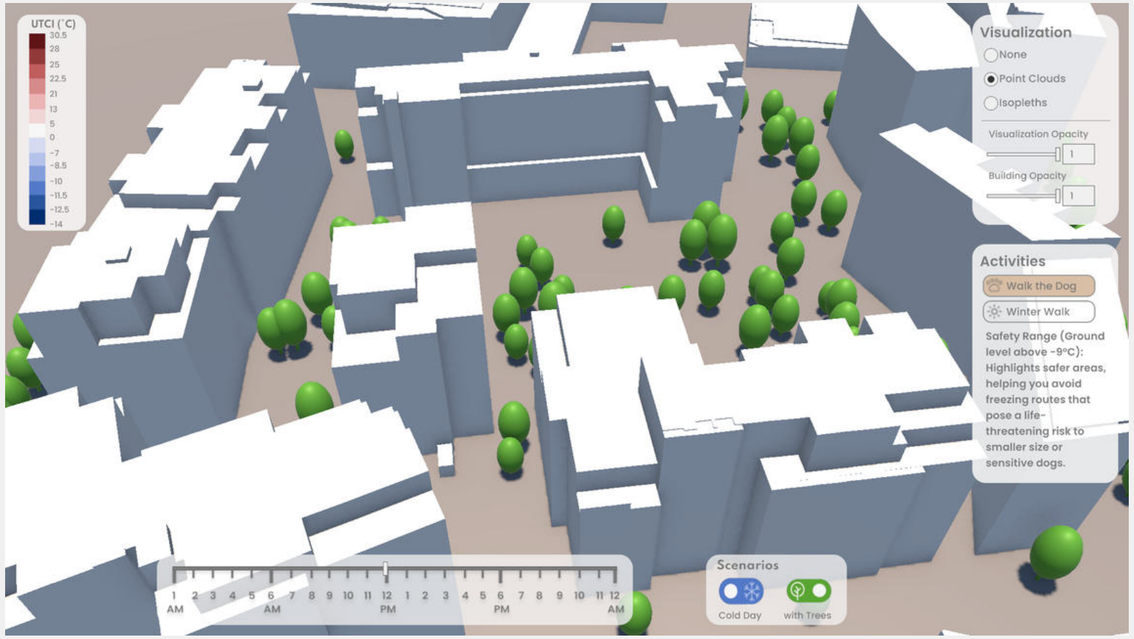


RESULTS AND DISCUSSION

User Frustration - Activity mode



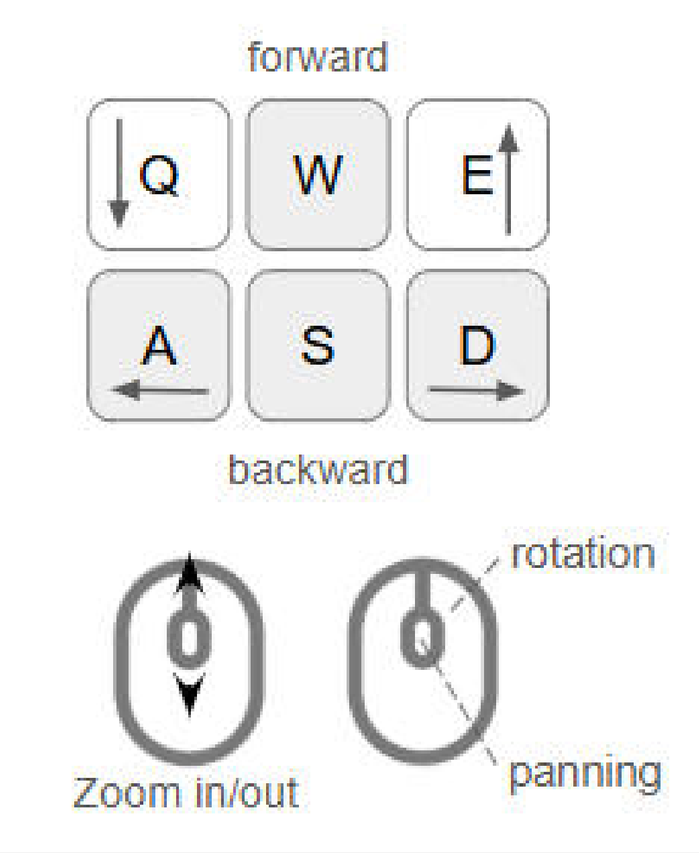
unclear visual logic



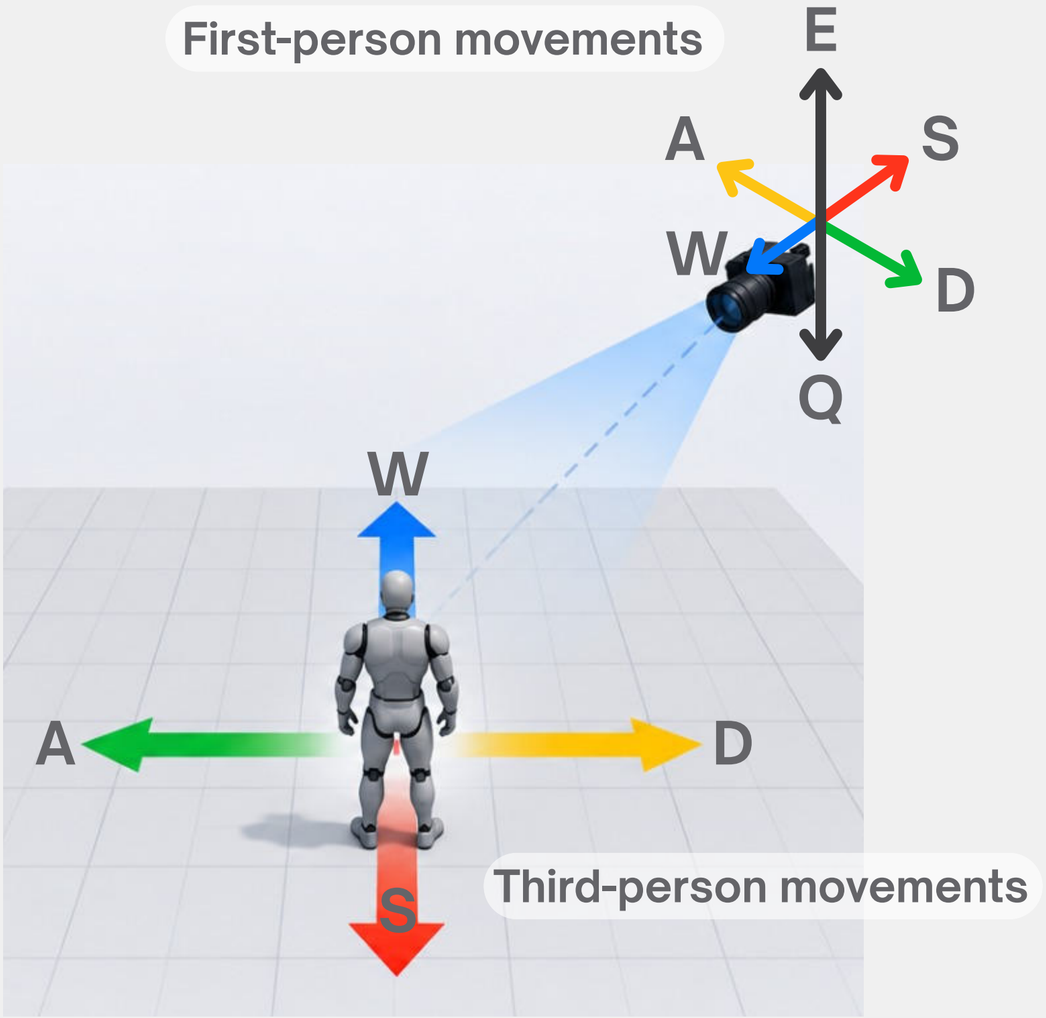
unclear empty-state feedback

RESULTS AND DISCUSSION

User Frustration - Camera navigation



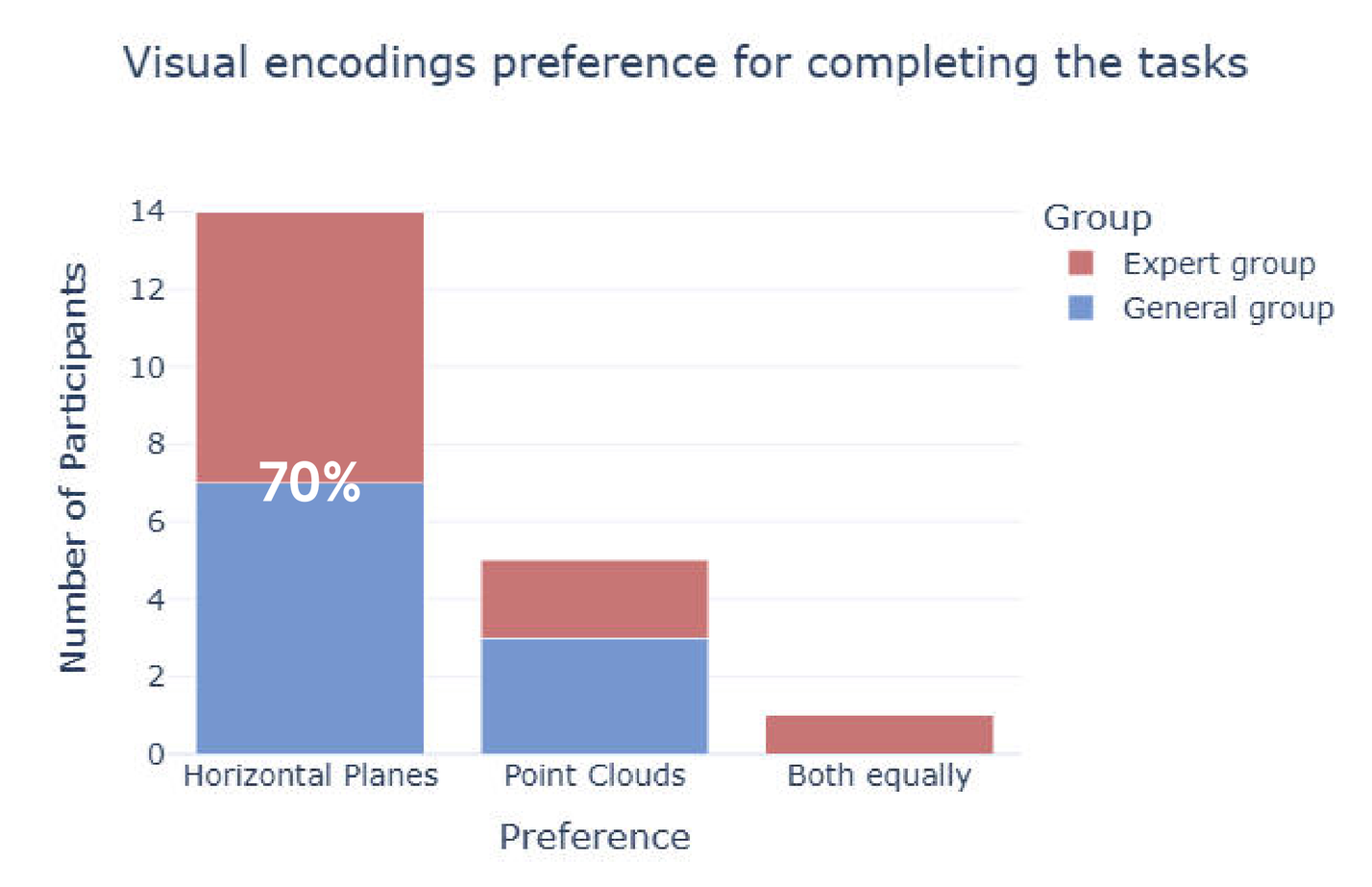
3D navigation controls in the prototype



AI-generated image; created with ChatGPT (OpenAI), based on author prompt, 2026.

RESULTS AND DISCUSSION

Visual encodings



RESULTS AND DISCUSSION

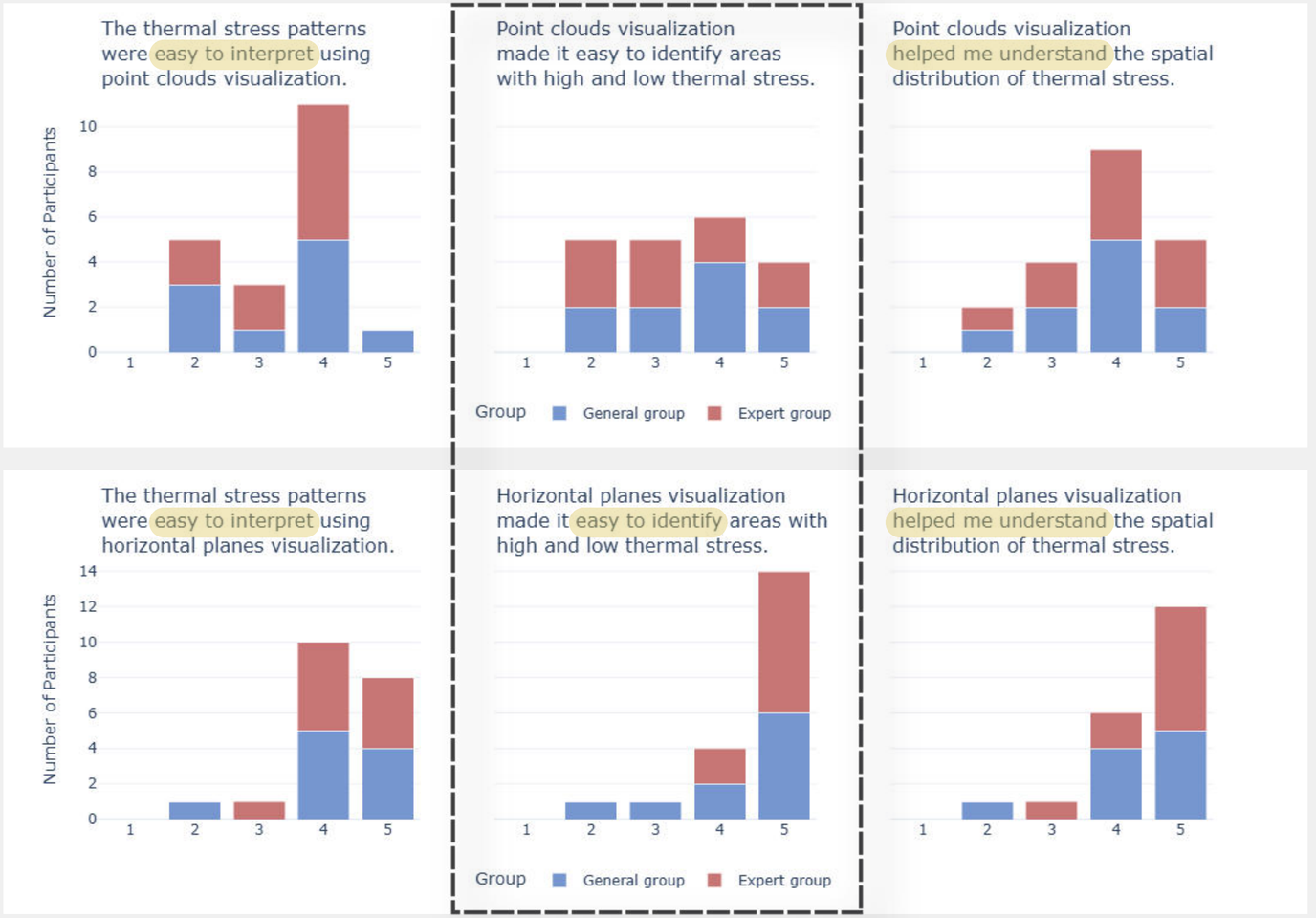
Visual encodings

Point clouds

- inspect local variations
- pedestrian eye-level

Horizontal planes

- rapidly identification
- bird eye view

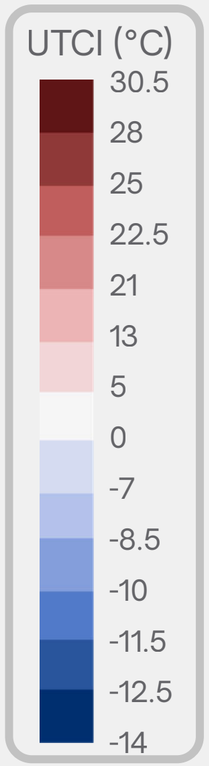


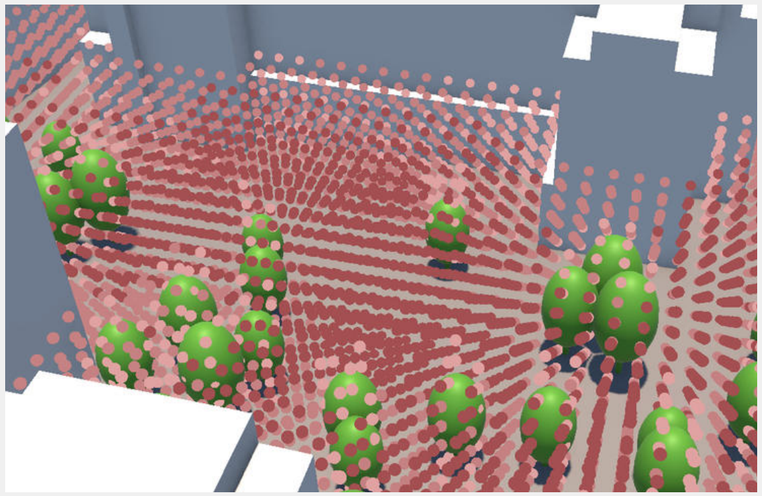
RESULTS AND DISCUSSION

Occlusion mitigation and transparency controls

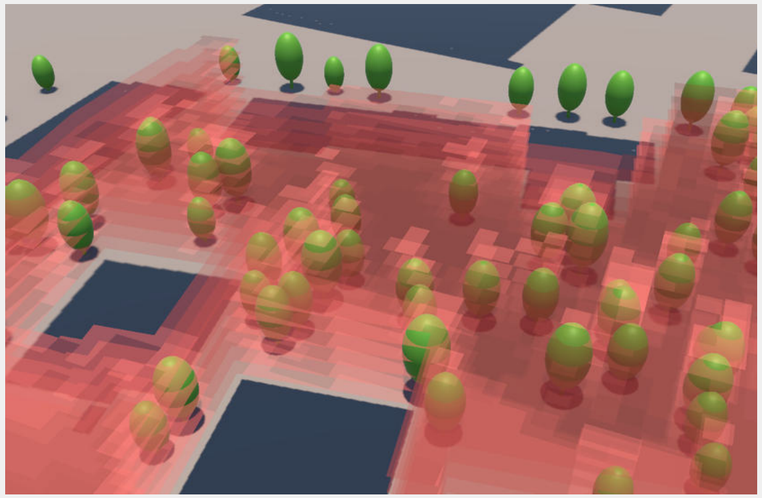
Opacity sliders are helpful!
However...

Mean ± std opacity slider usage time (second)		
	Building	Visualization
The locals	**11.33 ± 15.24**	**28.93 ± 24.27**
The experts	5.08 ± 6.92	16.78 ± 11.16





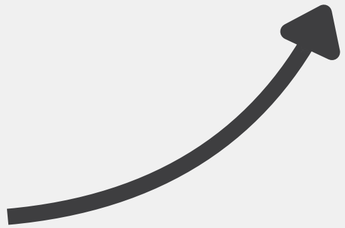
visual clutter



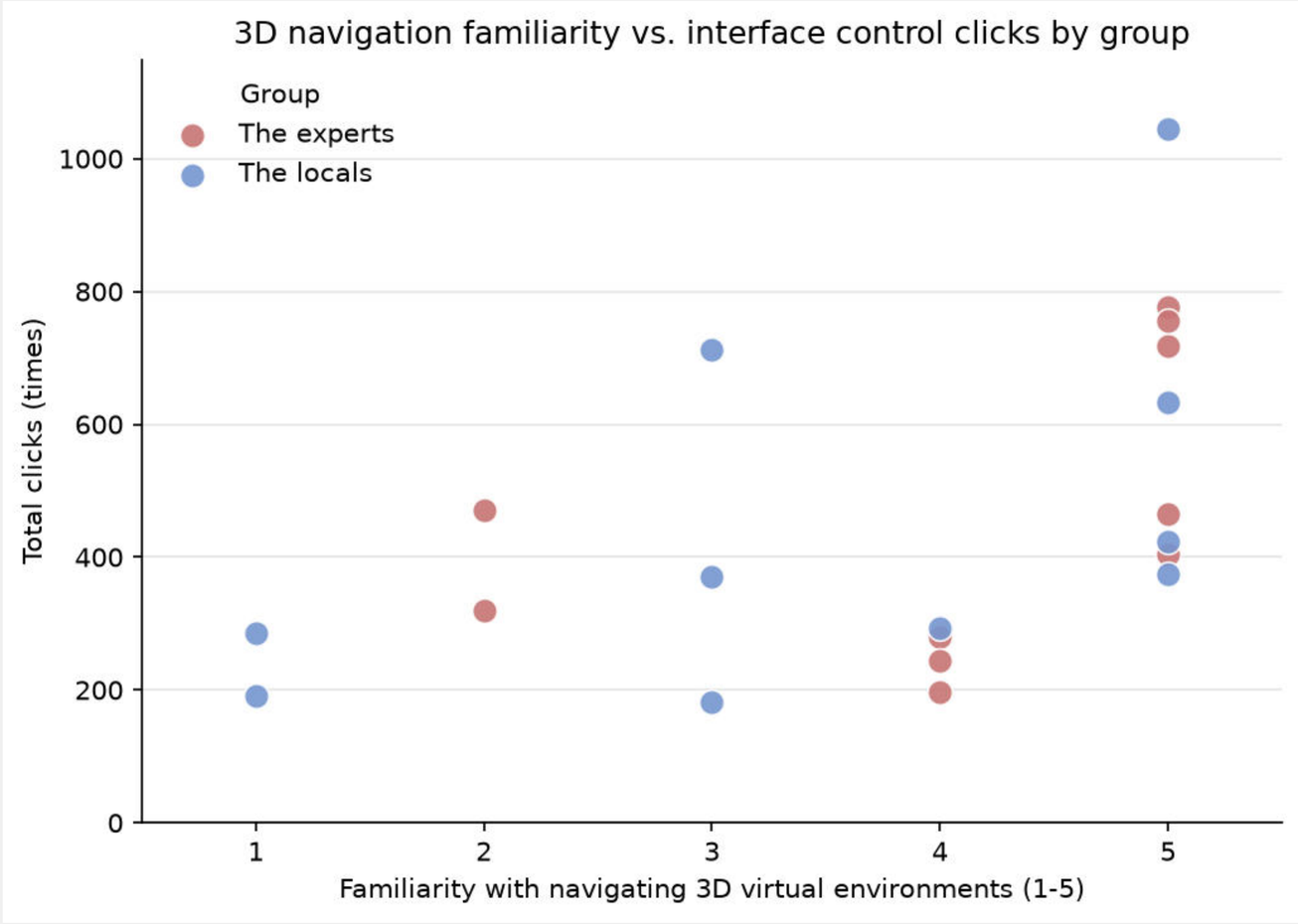
visual uncertainty

RESULTS AND DISCUSSION

Influence of user factors - 3D navigation familiarity

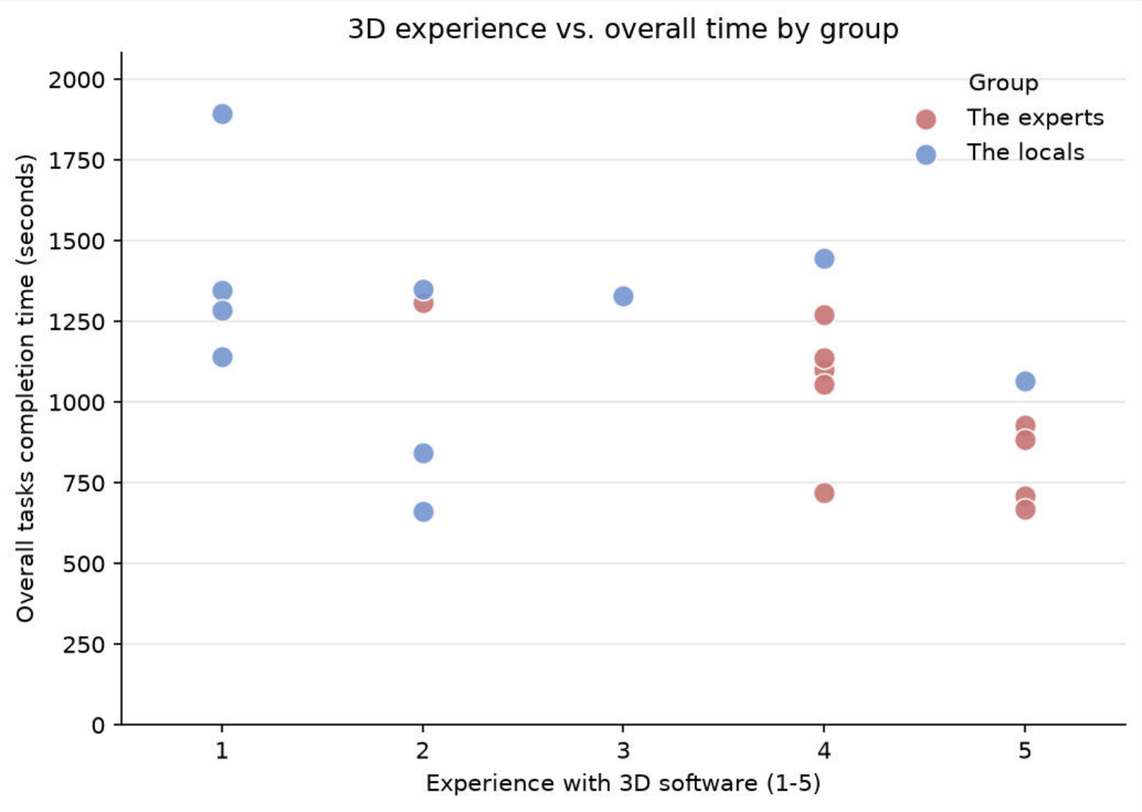


Positive trend

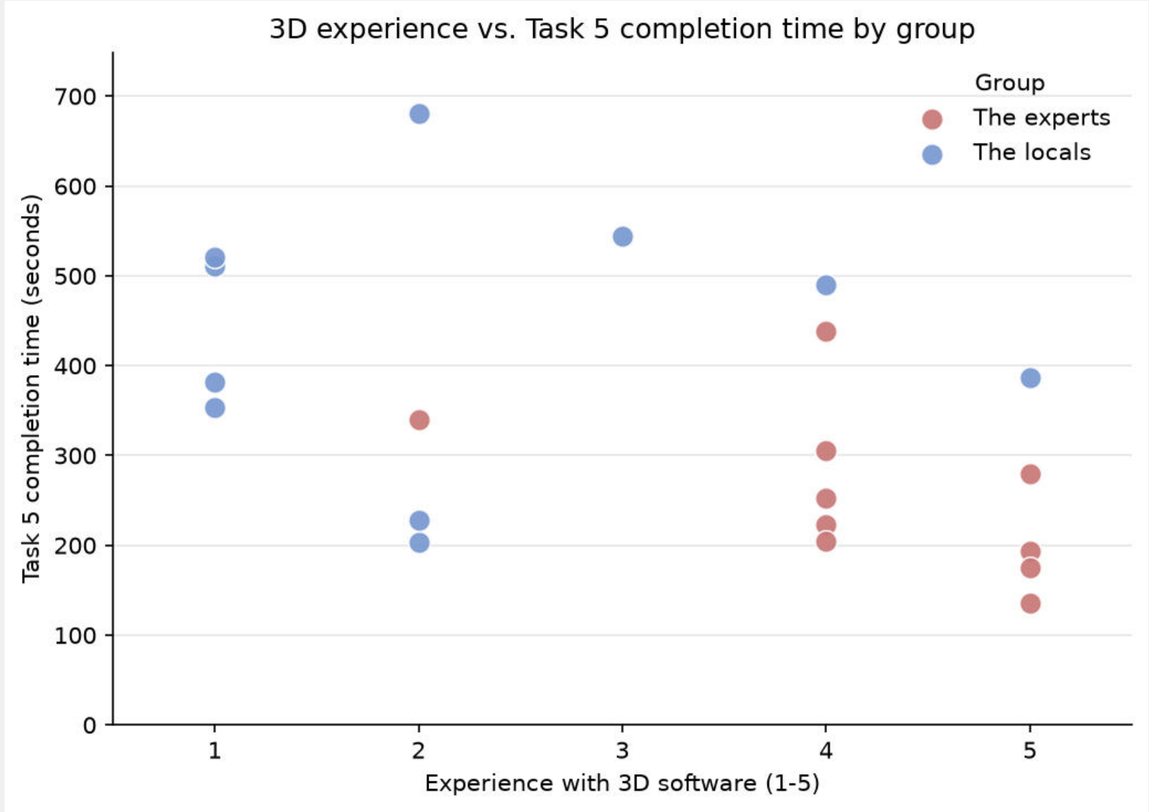


RESULTS AND DISCUSSION

Influence of user factors - 3D software experience



All tasks



Task 5 (most complex one)

KEY TAKEAWAYS

Viewpoint Matters

Horizontal planes → better from above

Point clouds → better at pedestrian level

Familiarity and Experience Matters

3D navigation familiarity → more exploration

3D software experience → faster task completion

REFERENCES

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THANK YOU

Any Questions or feedback?

Motivation 3

Research Objectives and Questions 4

Dataset and Simulation 7

Methodology 10

3D Prototype 11

User Study 22

Results and Discussion 27

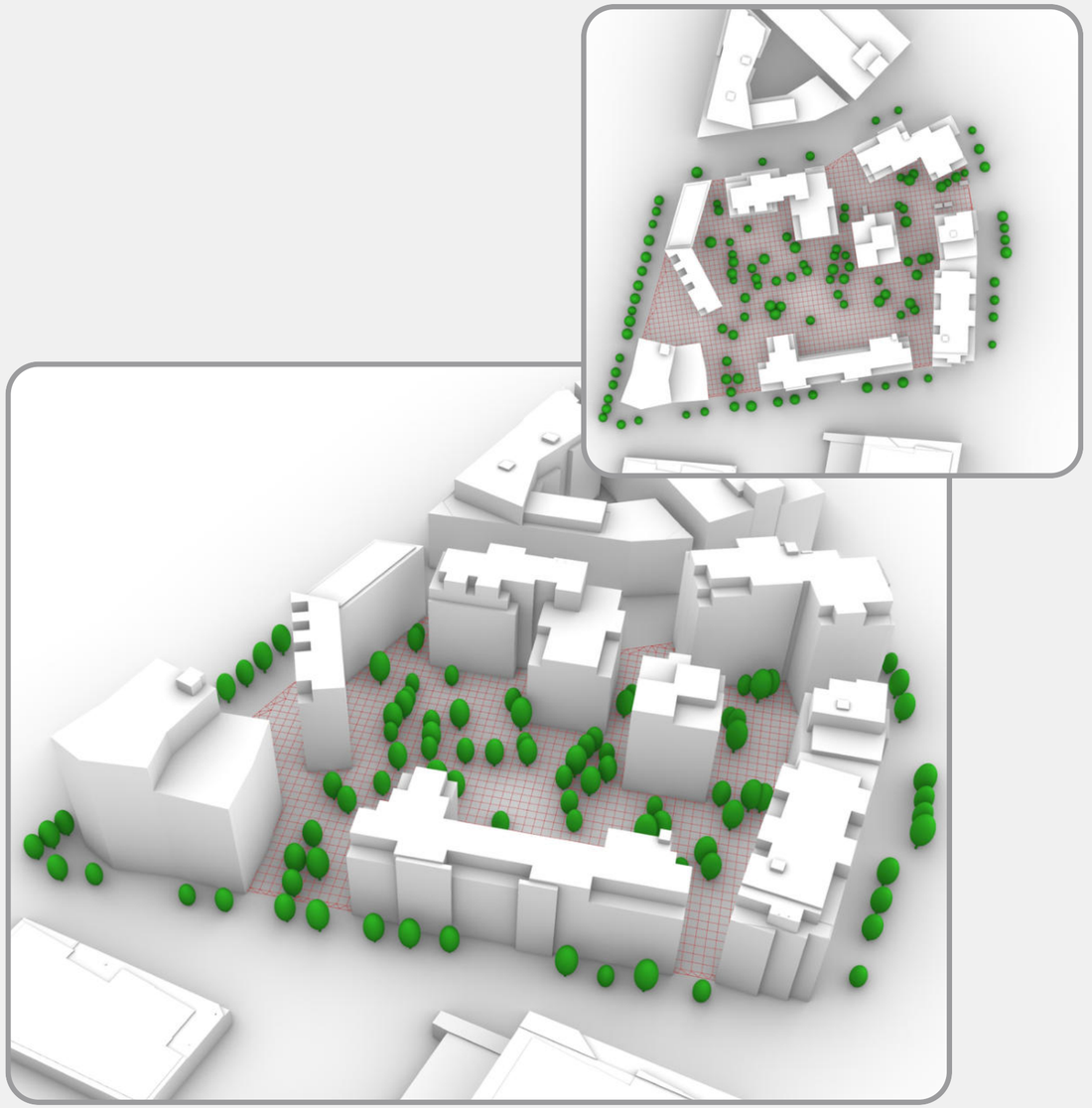
Key Takeaways 37

DATASET AND SIMULATION

Trees | Buildings | Ground Surface | Thermal Stress Indicator

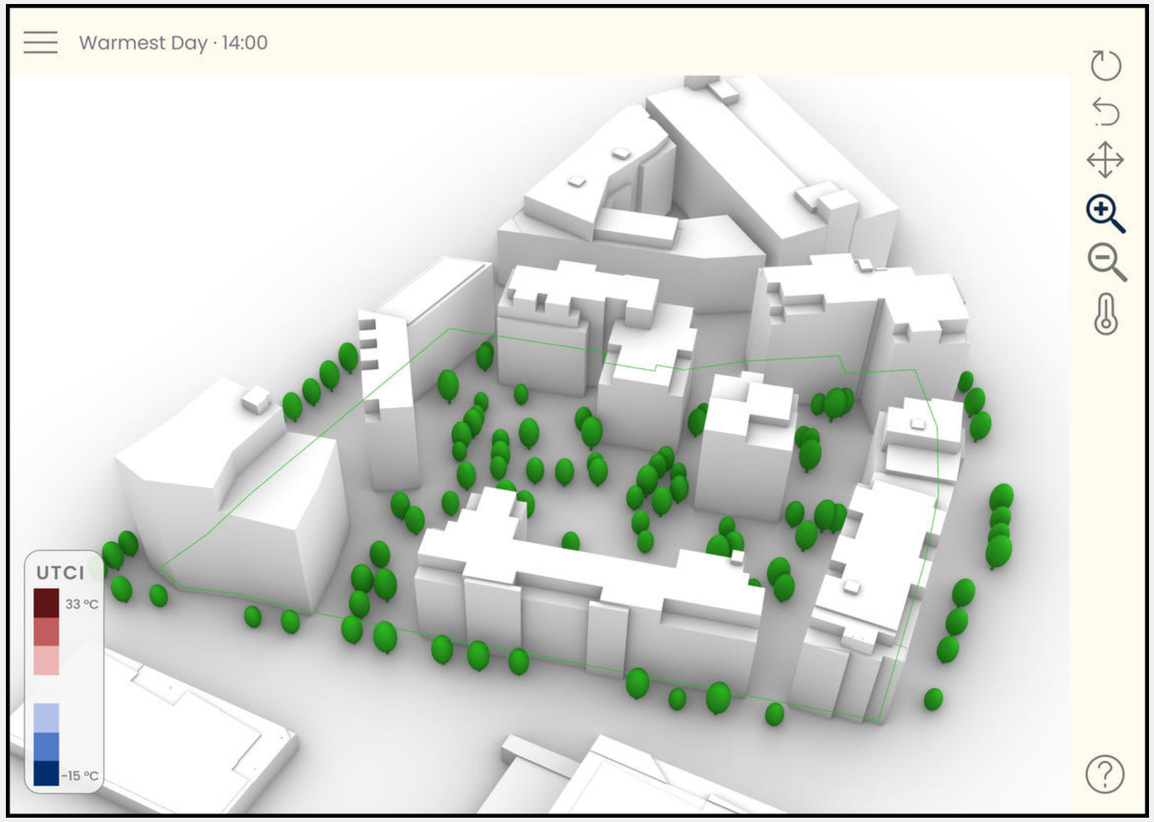
Simplifications during simulation:

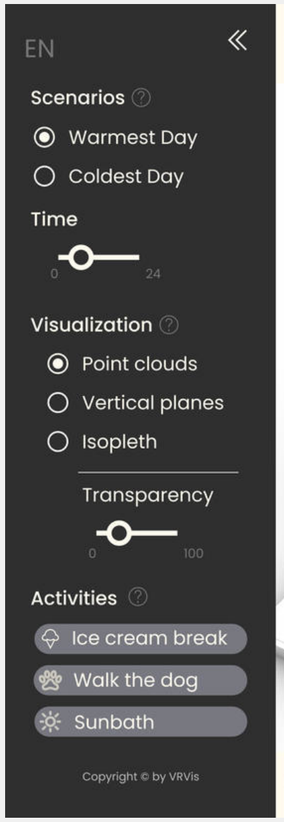
- The simulation mainly considered radiative forcing
- The trees were represented as simplified spheres that mainly function as shading objects
- The tree heights were assigned a randomized distribution

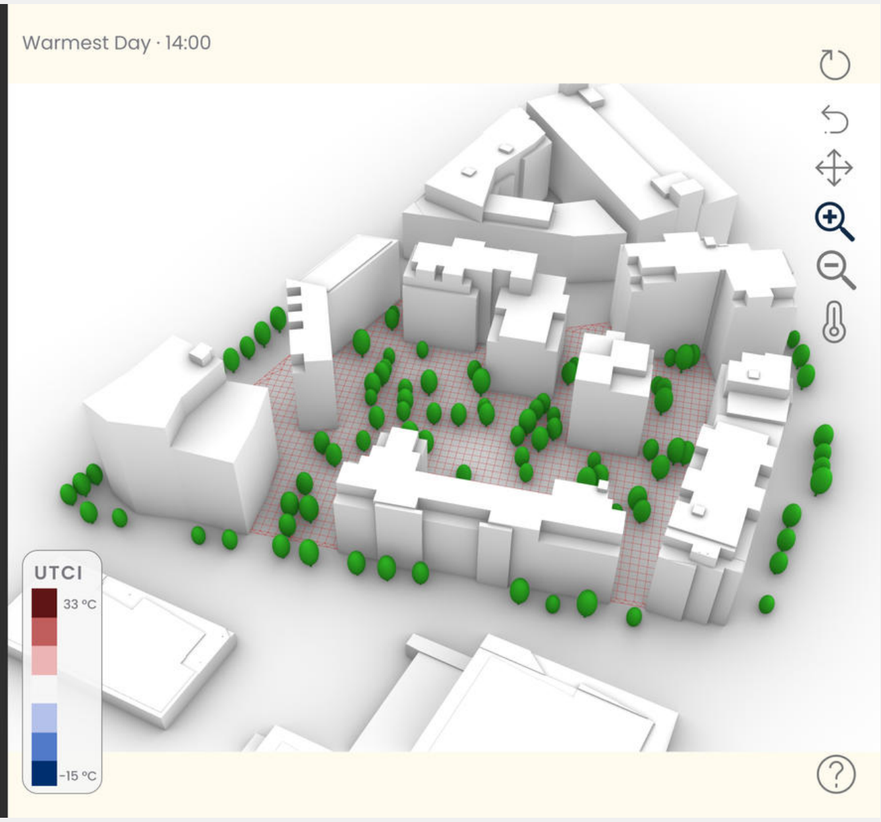


3D CARTOGRAPHIC PROTOTYPE

Mock-ups - 1st version

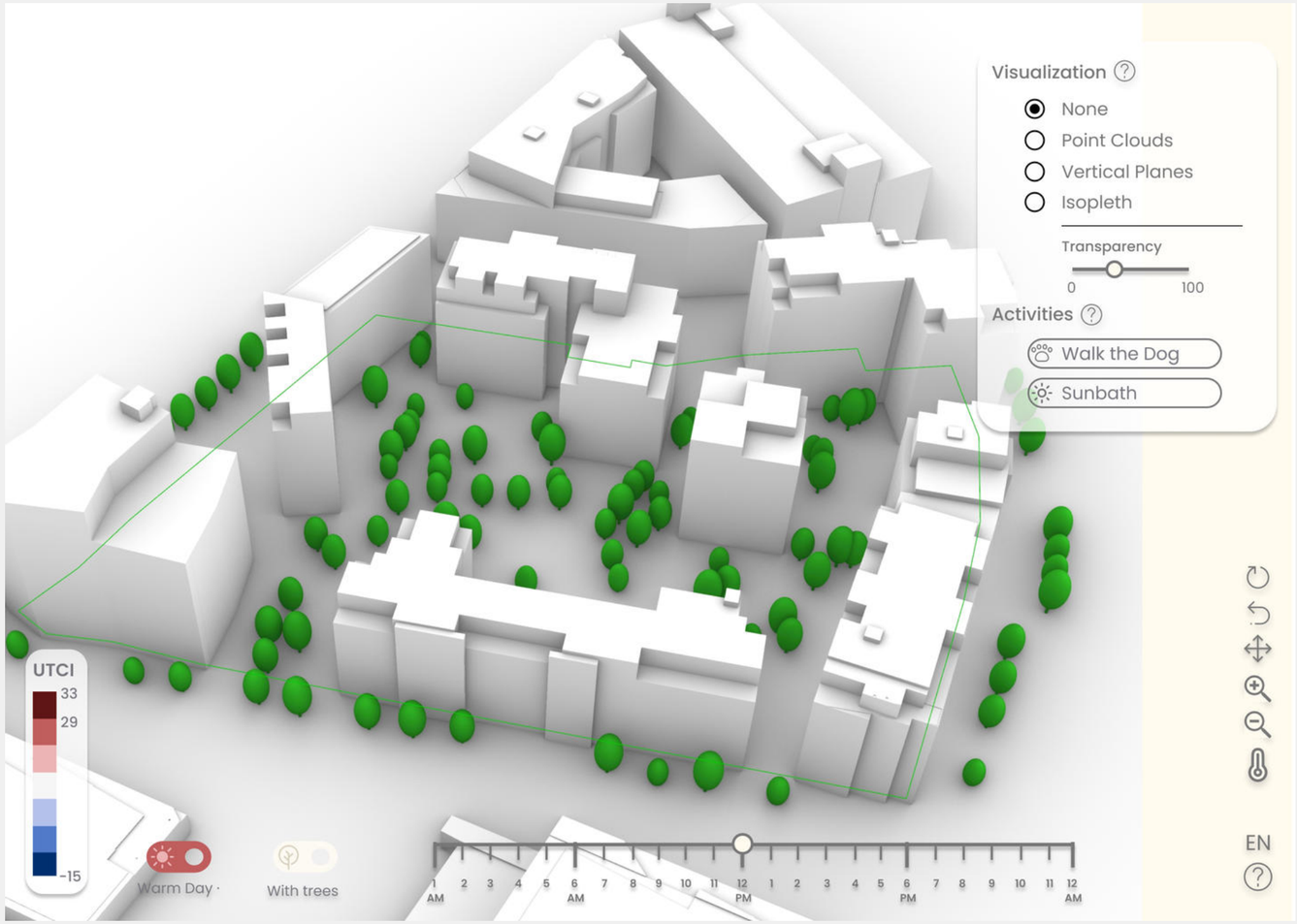






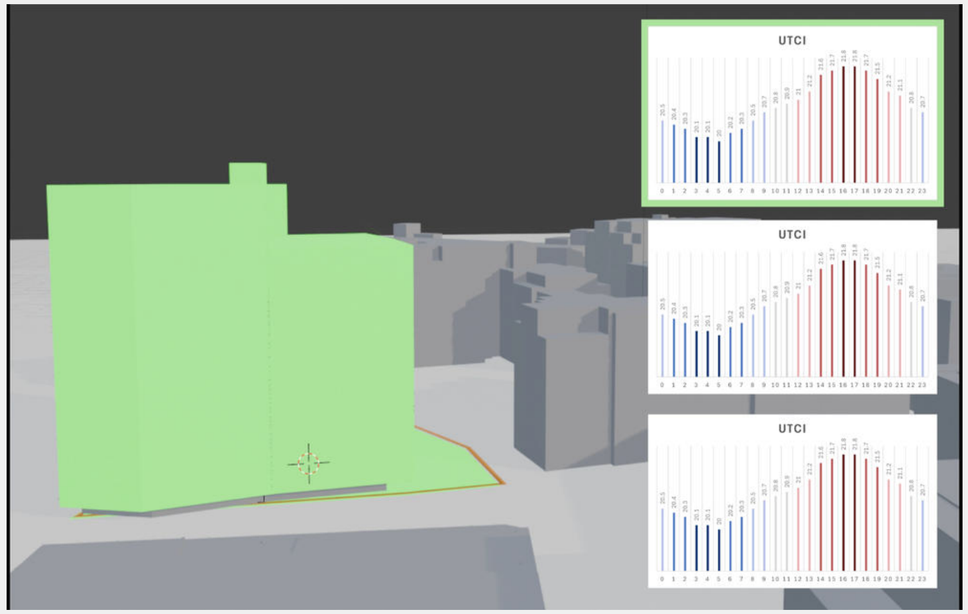
3D CARTOGRAPHIC PROTOTYPE

Mock-ups - 2nd version

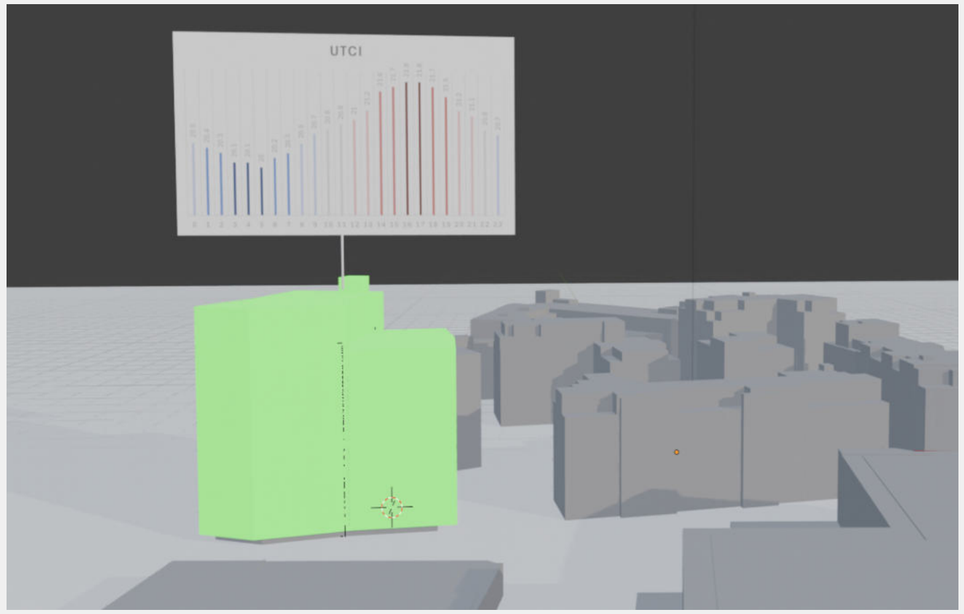


3D CARTOGRAPHIC PROTOTYPE

Analytical tools



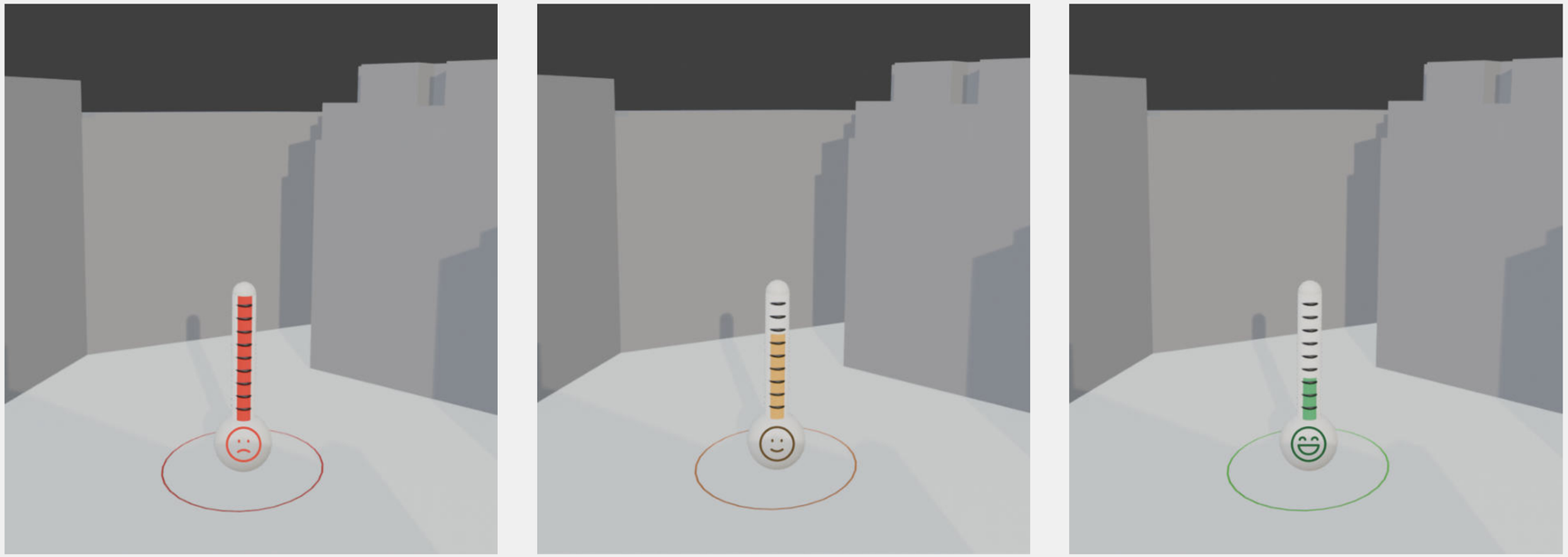
Linked View: Presents all zonal UTCI charts simultaneously in a sidebar (Mota et al., 2022).



Billboard View: Displays an hourly average UTCI chart positioned directly above a selected area (Mota et al., 2022).

3D CARTOGRAPHIC PROTOTYPE

Analytical tools



The mock-up of Thermal Stress Thermometer: Three states illustrate how the thermometer communicates thermal stress levels through color and facial expression (Rauer-Zechmeister, S. et al., 2024).

LIMITATIONS

Technical and Implementation Constraints

- Certain features, such as an thermal stress thermometer and 3D isopleths, could not be developed
- Virtual sunlight source was fixed directly overhead

User Study Design and Sample Distribution

- The sample size was insufficient for conducting statistically significant inferential analyses or formal correlation testing

Data Logging

- Interaction logs did not capture navigation metrics

FUTURE WORK

Address the identified limitations

- interactive height slider
- adaptive color schemes

Extend urban microclimate datasets beyond UTCI

- such as wind velocity, mean radiant temperature, and relative humidity

Investigate alternative platforms

- such as mobile devices, augmented reality (AR), virtual reality (VR), and CAVE